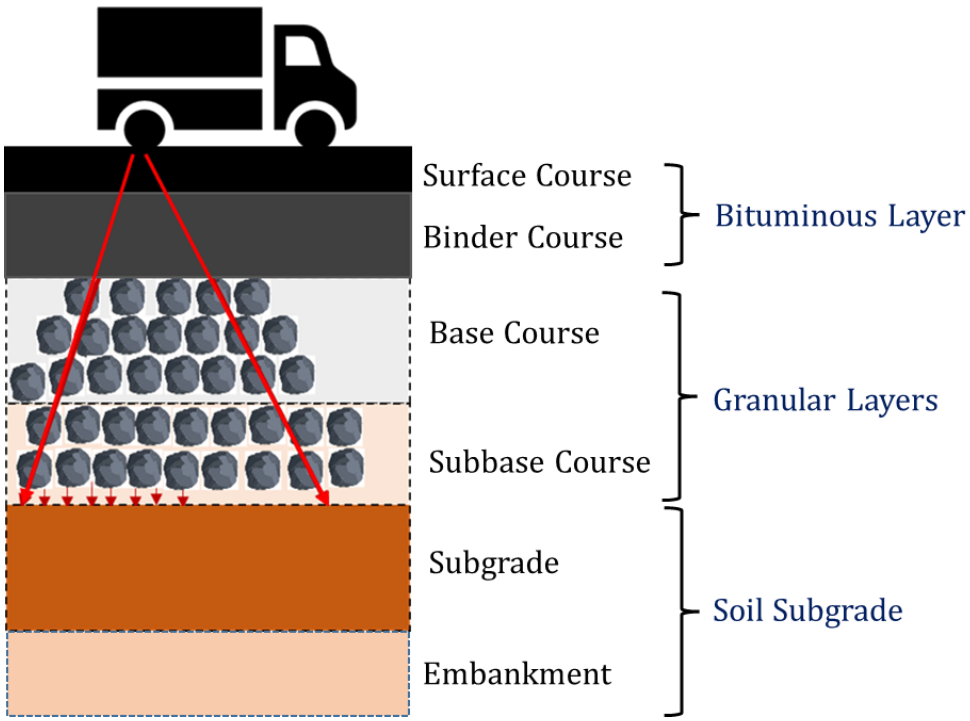
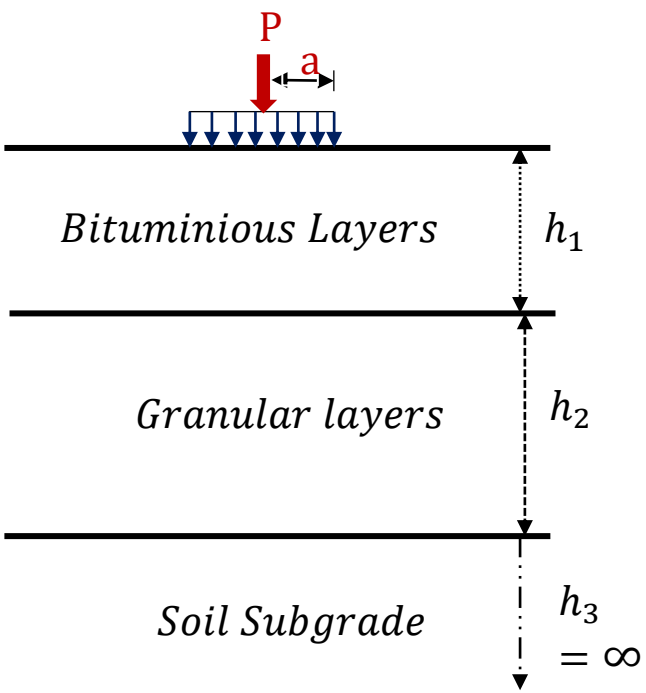
Design of Flexible Pavements





Dr. Surender Singh Email: surender@iitm.ac.in Phone: 9812955285 (M); 4313 (O)

Student Learning Outcome

At the end of this chapter, the students will be able to:

1. Identify the factors affecting the design of flexible pavements
2. Design the flexible pavements for Major Highways and for different conditions of loading and environment

Books/Codes Refereed for this unit:

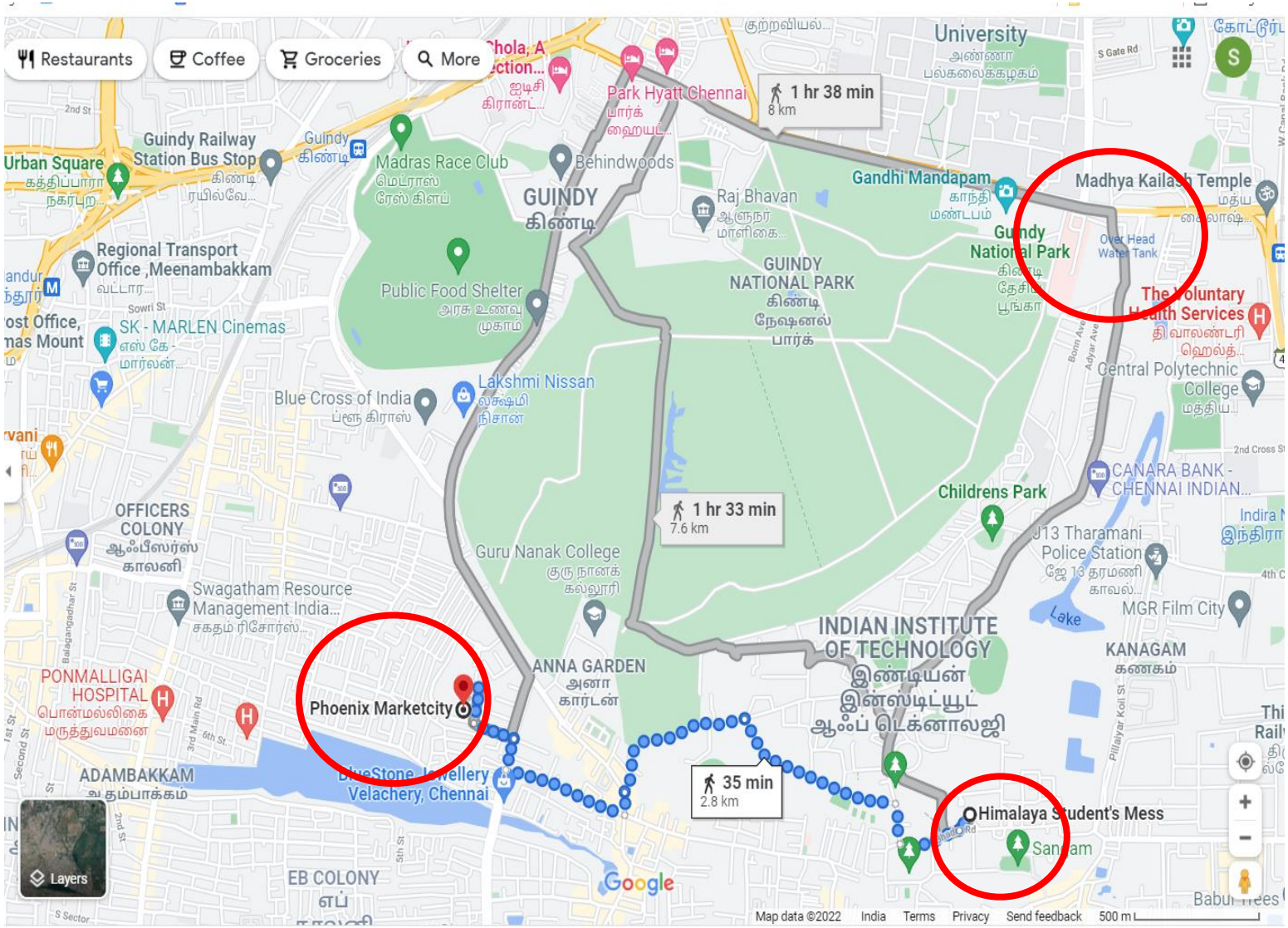
a) Pavement Analysis and Design, by Huang
b) IRC:37-2018
c) Principles of Pavement Design, by Yoder and Witczak

Challenge

Lets say you have to design a flexible pavement connecting Phoenix mall and Himalaya Canteen

What to design?

Lets say you have to design a flexible pavement connecting Phoenix mall and Himalaya Canteen



What to design?

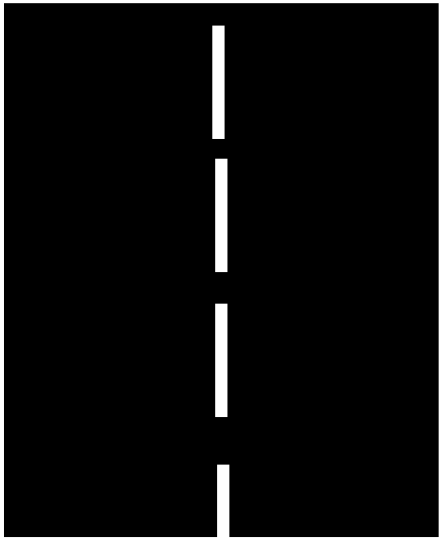
Design Parameters

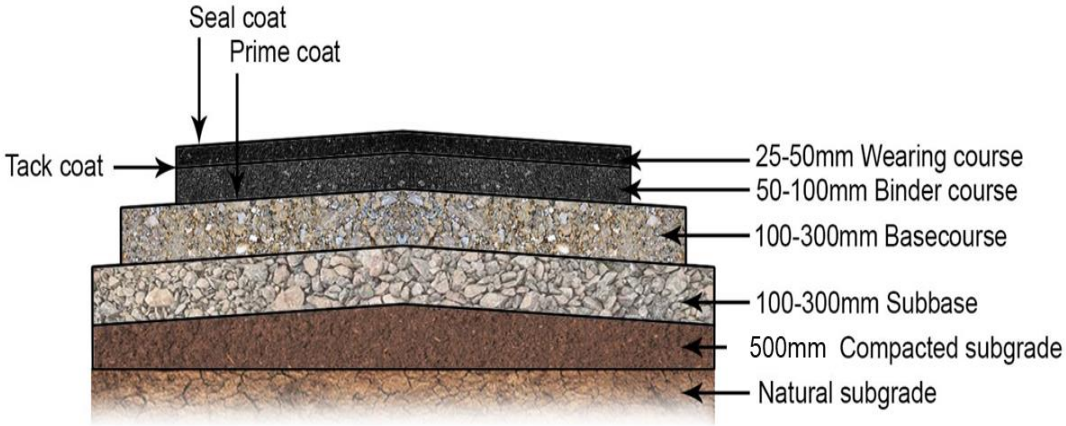
Geometrics
- Length
- Width
- Intersections
- Other furniture

Thickness
- Subgrade
- Sub-base & base
- Binder course
- Surface course

Materials
- Asphalt concrete
- Underlying layers
- Existing earth

Governs the Project ECONOMY as well as performance





Cross section of typical flexible pavement

How to design?

❖ Material Selection & Minimum Thickness

➢ Unbound Layers

- Subgrade: → 500 mm thick
- Subbase:
  - Granular Subbase (GSB)
    - Separation Layer → 100mm thick
    - Drainage Layer → 100mm thick
    - Combined Layer → 150mm thick
  - Cement treated Subbase (CTSB) → 200mm thick
- Base:
  - Granular (WBM, WMM, RCA)
    - If placed over GSB → 150mm thick
    - If placed over CTSB → 100mm thick
  - Cement Treated Base (CTB/cement treated-RCA) → 100mm thick
- Crack relief layer (over CTB)
  - Dense graded WMM → 100mm thick
  - Stress absorbing membrane interlayer (SAMI)
- Please remember that these are the minimum thickness and need to be checked for existing traffic condition (both during & post construction)

How to design?

❖ Material Selection & Minimum Thickness

➢ Bituminous Layers

- Binder Layer
  - Traffic > 20 MSA → Dense Bituminous Macadam (DBM)
  - Traffic < 20 MSA → Bituminous Macadam (BM)
- Surface Layer
  - Traffic > 20 MSA
    - Stone Matrix Asphalt (SMA)
    - Bituminous Concrete (BC)
    - Gap graded mix with rubberized bitumen (GGRB)
  - Traffic < 20 MSA
    - Bituminous Concrete (BC)
    - Semi Dense Bituminous Concrete (SDBC)
    - Pre-Mix Carpet (PMC)
    - Surface Dressing (SD)
    - Mix Seal Surfacing (SD)

- The thickness of these layers shall be determined based on the traffic conditions

Principles of Design

Design Approach

Empirical Design

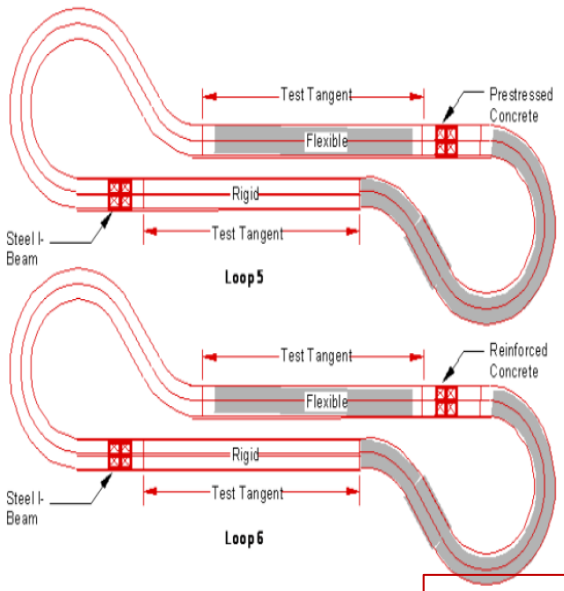
- Based on Field Performance, some regression equations will be generated
- Eg: AASHTO Design based on trials (1958-1960)

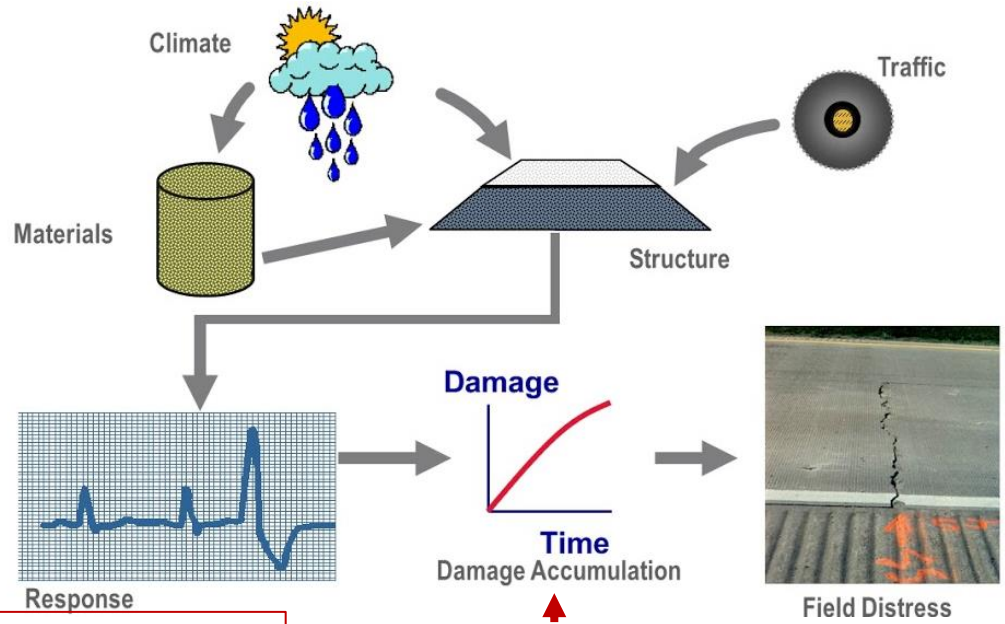
Mechanistic

- Based on mathematically calculated pavement response (FEM): Equations will developed

Mechanistic Empirical Design

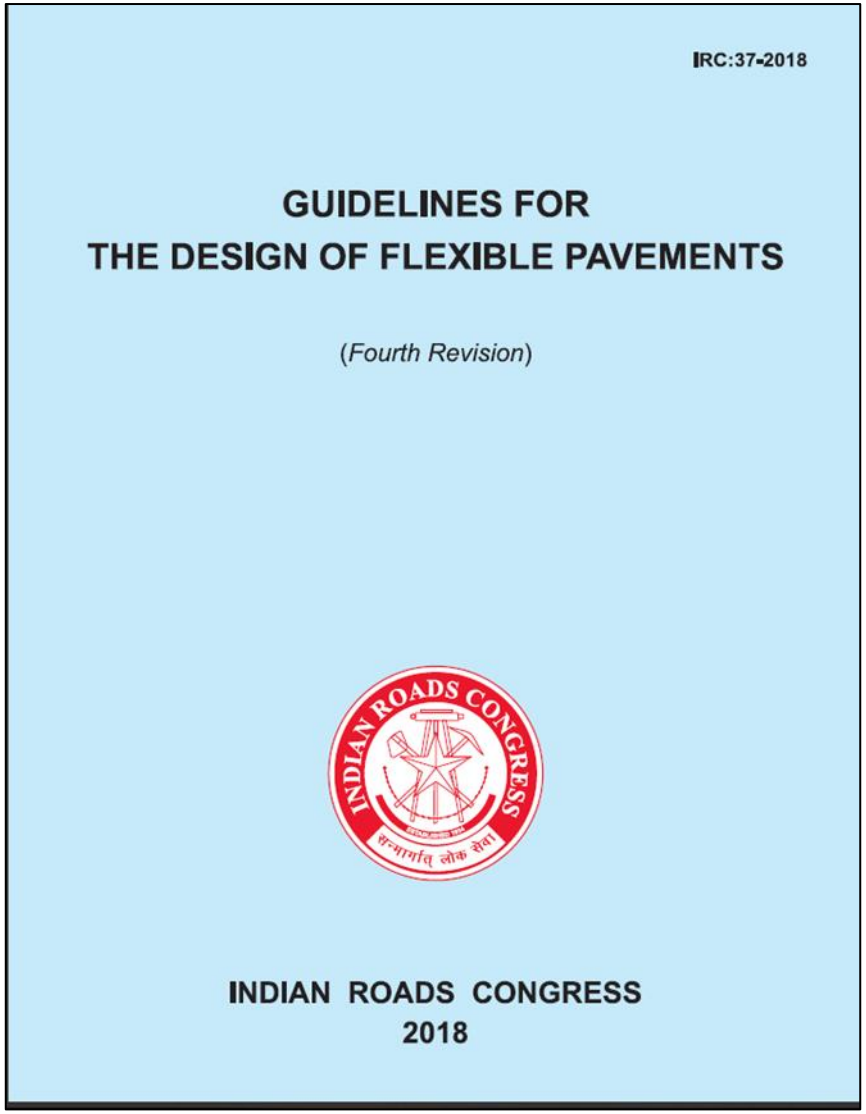
- Combination of both; considers empirical and mathematical approach
- Eg: IRC:37-2018





Data set from field studies

Mechanical-Empirical Design Procedure (MEPD): IRC:37



IRC:37-2018

GUIDELINES FOR
THE DESIGN OF FLEXIBLE PAVEMENTS

(*Fourth Revision*)

INDIAN ROADS CONGRESS
2018

Mechanical-Empirical Design Procedure (MEPD): Steps

Structural Model

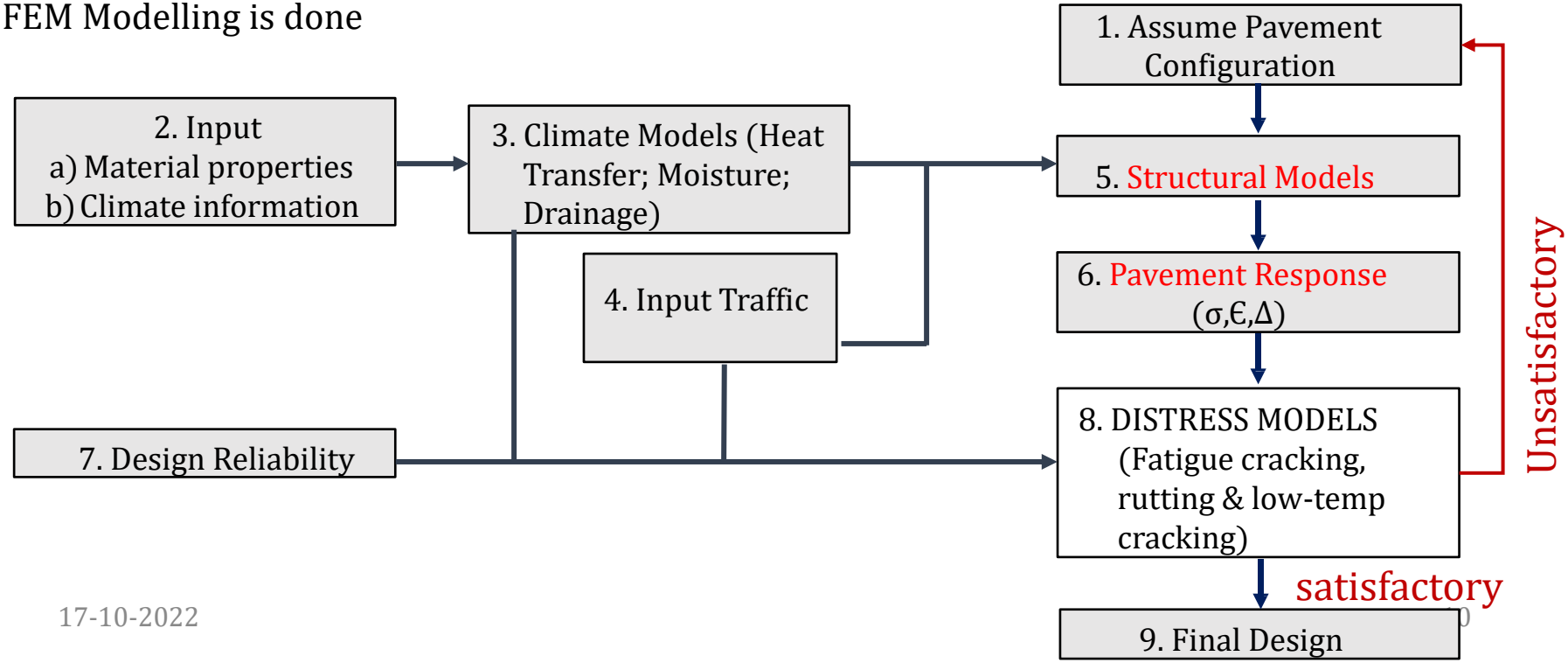
- Design the structure and predict the pavement response (stress & strain; deflection) with different input parameters
- FEM Modelling is done

Distress Models

- Predict the distress (failure) from pavement response
- Empirical regression eqn

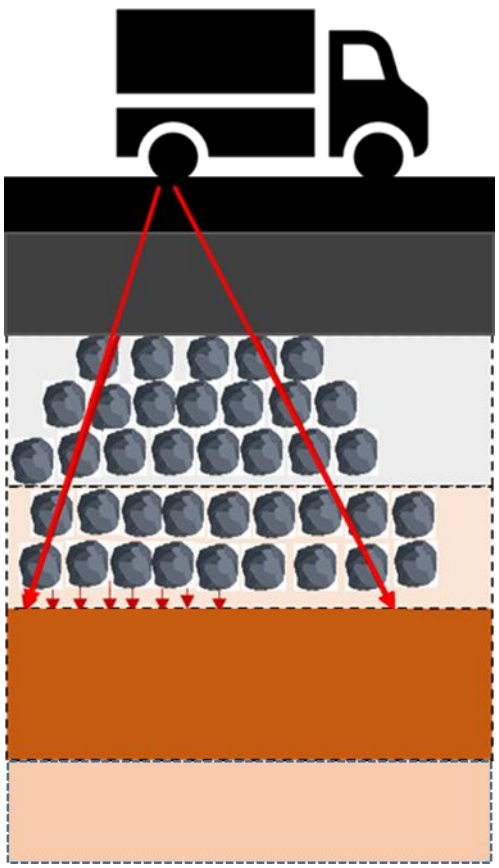
Calibration of Prediction with Field

- Calibrate the predicted distress with the field distress



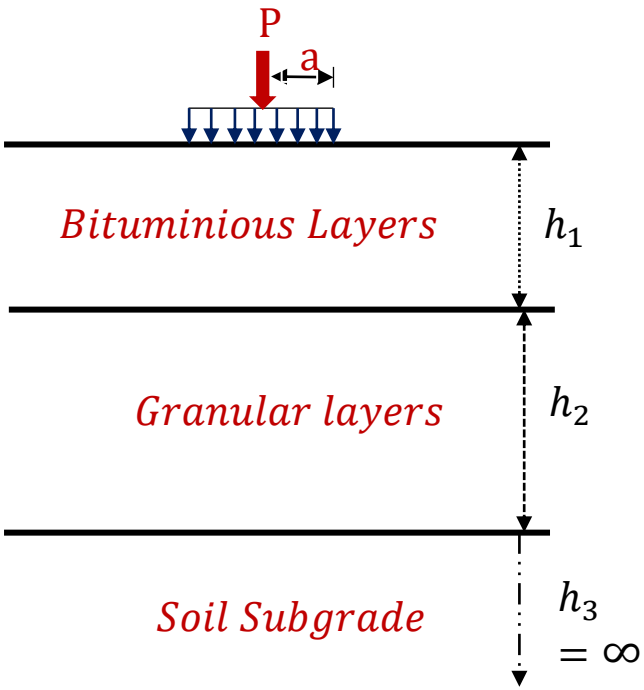
Structural Model Example considered in IRC:58-2015

- Structural Model: What should be components ? How will the system response?



Surface Course
Binder Course
Base Course
Subbase Course
Subgrade
Embankment

Bituminous Layer
Granular Layers
Soil Subgrade

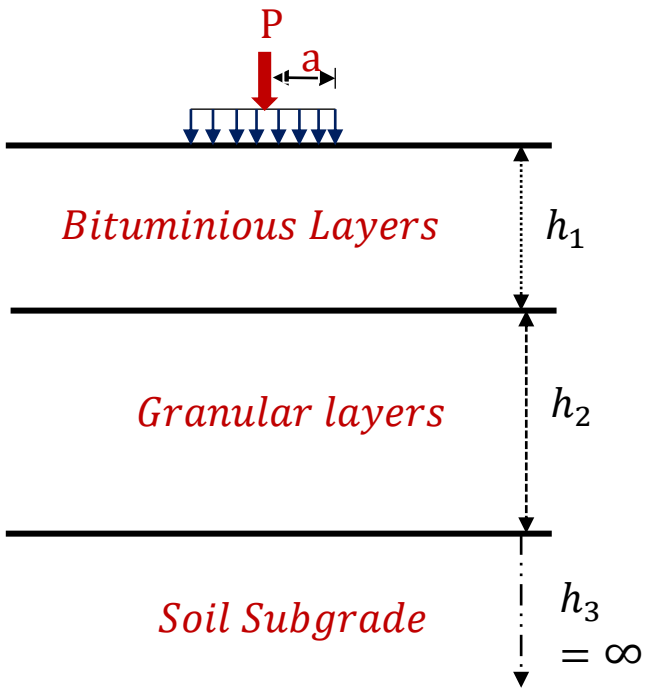


Actual Layers | For simplicity | Structural Model

- Let's discuss the parameters affecting the behavior of this structure…

Structural Model Example considered in IRC:58-2015

- Structural Model: What should be components ? How will the system response?



Parameters affecting the design

- What should be Load?
- Position of Load?
- Contact Pressure?
- Strength ?
- Response under load?
- Strength & response?
- Strength & response?

- Lets discuss these parameters in detail

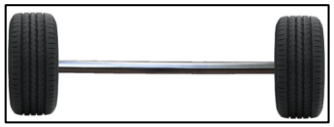
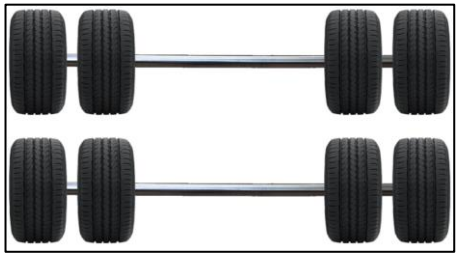
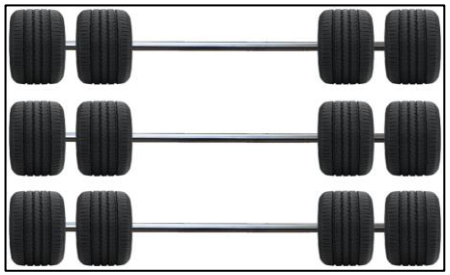
Design Components: Vehicle Load

❖ What is the basic difference in the following vehicles?

❖ Load carried and axle configuration

Design Components: Vehicle Load

- Contrary to rigid pavement design where the axles and their actual load is considered for thickness design, for flexible pavements, we design the pavement only for standard axles with standard loads

Axle Configuration	Description	Standard Load
	Single Axle with Single Wheel	65 kN
	Single Axle with Dual Wheels	80 kN
	Tandem Axle with Dual Wheels	148 kN
	Tridem Axle with Dual Wheels	224 kN

- What will happen if the axle carries more load than the standard axle load?

Design Components: Vehicle Load

- What will happen if the axle carries more load than the standard axle load?

- This will definitely lead to higher damage to the pavement compared to standard axle load and therefore, need to be accounted when designing the pavement

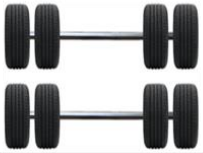
- This has been taken care in IRC:37 by considering the load equivalent factor called as Vehicle Damage Factor (VDF)

Design Components: Vehicle Damage Factor

- VDF is a multiplier for converting the number of commercial vehicles of different axle loads and axle configurations to the number of standard axle-load repetitions. It is defined as equivalent number of standard axles per commercial vehicle.

$$\text{VDF} = \left(\frac{Actual\ Axle\ Load\ in\ kN}{Standard\ Axle\ Load\ in\ kN}\right)^4$$

- Forth Power law could be used to determine the equivalent damage by an axle with respect to standard axle

Standard Axle	Standard Load	Axle Configuration	VDF
Single axle with SINGLE Wheels	65 kN		$\left(\frac{Actual\ Axle\ Load}{65}\right)^4$
Single axle with DUAL Wheels	80 kN		$\left(\frac{Actual\ Axle\ Load}{80}\right)^4$
Tandem Axle with Dual Wheels	148 kN		$\left(\frac{Actual\ Axle\ Load}{148}\right)^4$
Tridem Axle with Dual Wheels	224 kN		$\left(\frac{Actual\ Axle\ Load}{224}\right)^4$

Design Components: Vehicle Damage Factor

- For a highway to be designed, we have to calculate the average VDF (average of all the vehicles) in the predominant direction
- Calculate the average VDF for the below traffic

S.No.	Axle Load, kN			
	Single Axle Bus		Tandem axle truck	
	Front axle	Rear axle	Front axle	Rear axle
1	22.06	44.13	40.50	162
2	22.88	45.76	41.48	165.92
3	23.53	47.07	49.62	198.47
4	38.41	76.81	55.70	222.79
5	40.20	77.47	57.17	228.68
6	40.37	80.41	61.39	245.54

- Front axles are always single axles with single wheels on both sides $\text{VDF} = \left(\frac{Actual\ Axle\ Load\ in\ kN}{65}\right)^4$
- Rear axles are single with dual wheels for bus & tandem for trucks $\text{VDF} = \left(\frac{Actual\ Axle\ Load\ in\ kN}{80/148}\right)^4$

Design Components: Vehicle Damage Factor

$$\text{VDF} = \left(\frac{Actual\ Axle\ Load\ in\ kN}{65}\right)^4 \qquad \text{VDF} = \left(\frac{Actual\ Axle\ Load\ in\ kN}{80}\right)^4$$

S.No.	Axle Load		Damaging effect		
	Front axle	Rear axle	Front Axle	Rear Axle	Total
1	22.06	44.13	0.013	0.093	0.106
2	22.88	45.76	0.015	0.107	0.122
3	23.53	47.07	0.017	0.120	0.137
4	38.41	76.81	0.122	0.850	0.972
5	40.20	77.47	0.126	0.879	1.005
6	40.37	80.41	0.146	1.021	1.167
Total					3.509
Average VDF					0.585

Design Components: Vehicle Damage Factor

$$\text{VDF} = \left(\frac{Actual\ Axle\ Load\ in\ kN}{65}\right)^4 \qquad \text{VDF} = \left(\frac{Actual\ Axle\ Load\ in\ kN}{148}\right)^4$$

S.No.	Axle Load		Damaging effect		
	Front axle	Rear axle	Front Axle	Rear Axle	Total
1	40.50	162	0.151	1.435	1.586
2	41.48	165.92	0.166	1.580	1.745
3	49.62	198.47	0.340	3.234	3.574
4	55.70	222.79	0.539	5.135	5.674
5	57.17	228.68	0.598	5.699	6.298
6	61.39	245.54	0.795	7.576	8.372
Total VDF					27.249
Average VDF (Total/no of trucks)					4.542

Design Components: Vehicle Damage Factor

- Weighted Average VDF $= \frac{C_1(VDF1)+C_2(VDF2)+C_3(VDF3)\ldots}{C_1+C_2+C_3+\cdots}$

  - $C_x = no\ of\ vehicles\ count$

Vehicle Type	VDF	Count	VDF X count
Bus	0.947	109	103.223
Tandem Truck	3.127	70	218.89
Tridem Truck	7.121	112	797.552
LCV	0.181	123	22.263
Total		414	1081.928
Weighted Average VDF			2.61

Design Components: Vehicle Damage Factor

- Indicative Vehicle Damage Factors for small projects

- In absence of the axle load distribution (small projects), the VDF values could be adopted from the following table

CVPD (2-way)	Terrain	
	Rolling/Plain	Hilly
0-150	1.7	0.6
150-1500	3.9	1.7
>1500	5.0	2.8

Design Components: Critical Load Position

❖ Which is the critical position under loading?

- There is no concept of critical position and therefore, we have to design the pavement for all the vehicles plying on the road. However, lateral distribution of the vehicles over the carriageway need to be considered for realistic design; else it will be too expensive

- Lane Distribution Factor (D): It is the lateral distribution of vehicles, and this depends on the type of the road

Undivided Roads

Divided Roads

❖ For UNdivided roads, we have to consider total traffic i.e. the lateral distribution is higher

❖ For divided roads, we have to consider only the traffic in the predominant direction , i.e. lower lane distribution and thus low damage

Design Components: Critical Load Position

- Values of Lane Distribution Factor (D)

Undivided Carriageways/Roads

Total Lanes	D
Single Lane Roads	100%
2 Lane Single Carriageway	50%
4 Lane Single Carriageway	40%

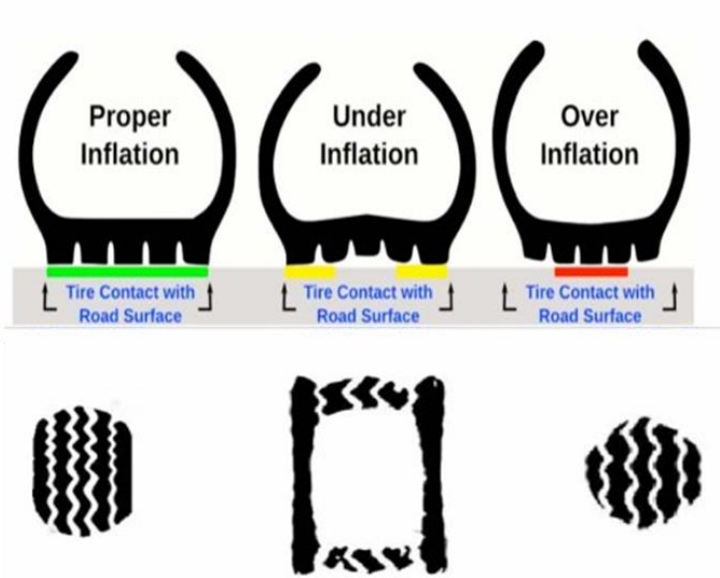
Divided Carriageways/Roads

Lanes in each direction	D
Two lanes	75%
Three Lanes	60%
Four Lanes	45%

Design Components: Tire Pavement Contact

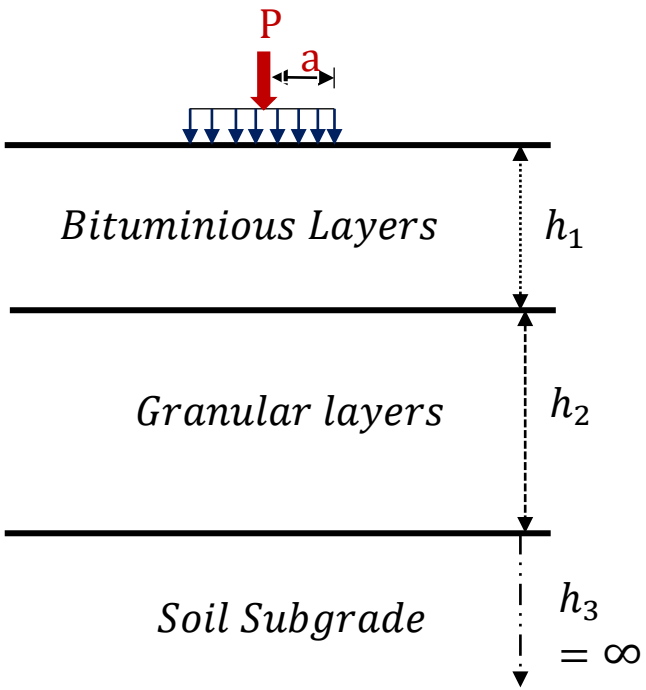
❖ Tire-Pavement Interaction

- In the mechanistic design, it is necessary to know the contact area so that axle load can be assumed to uniformly distributed over this area
- Contact Pressure $= \frac{Load\ on\ Wheel}{Contact\ Area}$
- The contact Area is also dependent on the Tire pressure
- Higher tire pressure is detrimental to the pavement surface (erosion of surface) but not to the structure integrity of pavement
- Therefore, a contact pressure value of 0.56 MPa has been considered except for CTB (0.8 MPa as in concrete pavement design)



Structural Model Example considered in IRC:58-2015

❖ Structural Model: What should be components ? How will the system response?



Parameters affecting the design

- What should be Load? Adjusted Standard loads using VDF
- Position of Load? Lane distribution factor
- Contact Pressure? 0.56 MPa
- Strength ?
- Response

- Strength & response?

- Strength & response?

❖ We can discuss strength & Response together

Layered Elastic Model

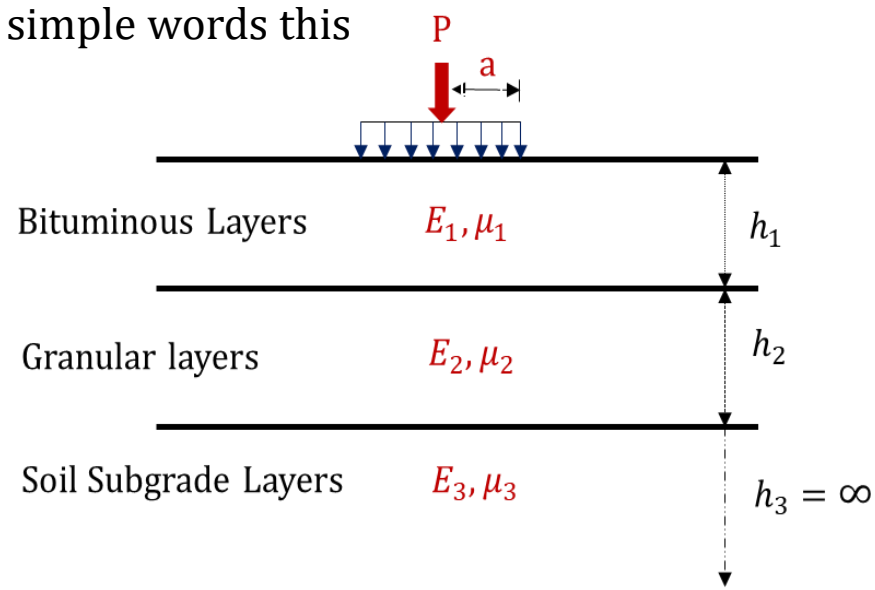
❖ Analysis of Stresses in Flexible Pavements using Layered Elastic model

- A layered is a Mechanistic Model (computer operated)
- It was proposed by V.J. Boussinesq in 1885
- It can compute stresses, strains and deflections at any point in a pavement structure resulting from the application of a surface load

➢ Basic assumption is that the pavement structural layer is homogeneous, isotropic, and linearly elastic. In simple words this means the following:

i. The response is same everywhere in a particular layer and the pavement will rebound to its original form once the load is removed

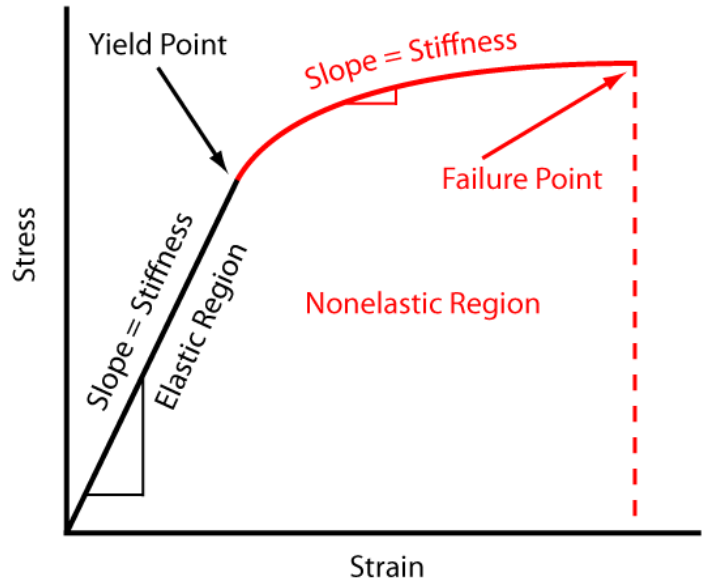
ii. pavement layers can be characterized by two constants - Young's modulus E and Poisson's ratio μ



P is the wheel load, a is the radius of circular tire imprint

Strength Characteristics of Layers

❖ Elastic Modulus & Poisson's ratio



- Elastic Modulus (E) $= \frac{Stress}{Strain}$ (upto Elastic Limit)
- Poisson's Ratio $(\mu) = \frac{Longitudinal\ Strain}{Lateral\ Strain}$
- Strength $= \frac{Ultimate\ Load\ at\ Failure}{Area\ of\ the\ loaded\ specimen}$

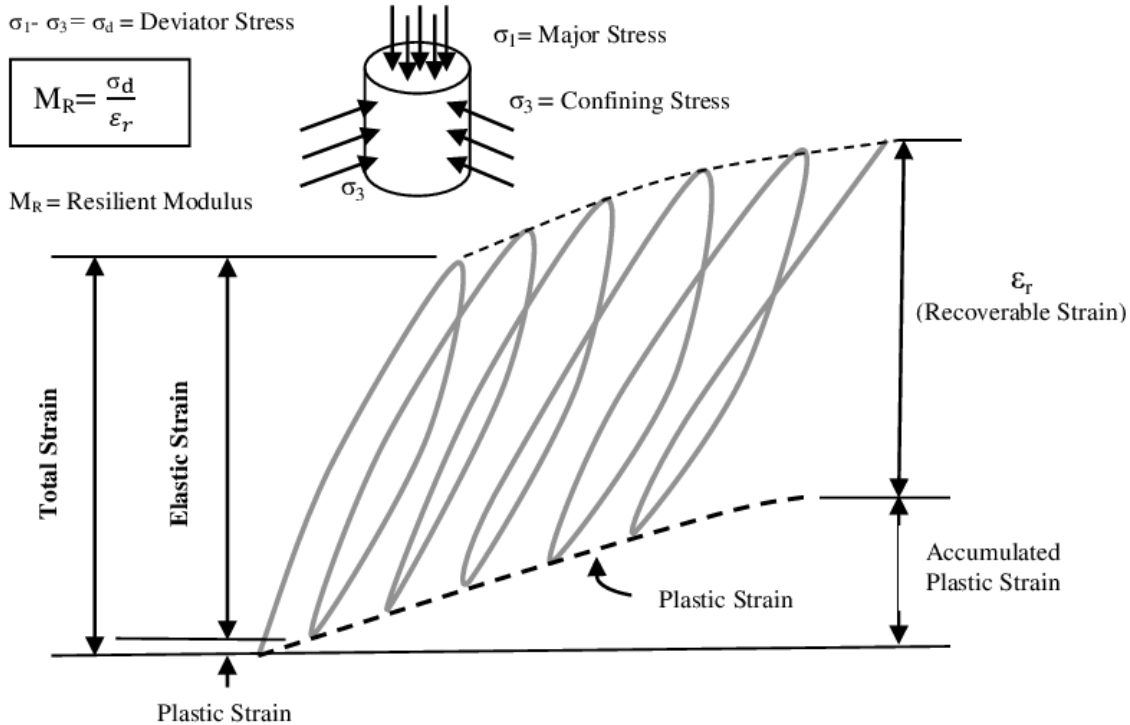
➢ However, resilient modulus (M_R) is used instead of elastic modulus (E) for pavement applications due to :

- ✓ to account repeated/cyclic rapid applied loads (in E, slow and constant load is applied which does not simulate actual vehicle movement)
- ✓ To account permanent deformation (strain) in the unbound layers during the construction (in E, only elastic strain is considered)

Strength Characteristics of Layers

- M_R is the elastic modulus of layers based on Recoverable strain under cyclic/repeated loading

$$M_R = \frac{\text{Repeated Axial Stress}}{\text{Recoverable Strain}}$$



- However, conducting resilient modulus (M_R) needs sophisticated & expensive machines and therefore, empirical models or indicative values are used to evaluate the same

Empirical Models to Estimate M_R

- Resilient Modulus of Subgrade (M_{RS})

$$M_{RS} = 10.0 \times CBR \text{ for } CBR \leq 5\ \%$$

$$M_{RS} = 17.6 \times (CBR)^{0.64} \text{ for } CBR > 5\ \%$$

- Resilient Modulus of Granular Layers (M_{RGRAN})

$$MR_{GRAN} = 0.2\ (h)^{0.45} \times M_{RS}$$

where: 'h' is the thickness of granular layers in mm

- Resilient Modulus of Cement Treated Subbase (M_{RCTSB})

$$E = 1000 \times UCS$$

where: UCS is the unconfined compressive strength (MPa) at 28 days of curing

- Resilient Modulus of Bituminous Layers could be selected from the table

Response of Structure using IITPAVE

❖ Resilient Modulus & Poisson's ratio

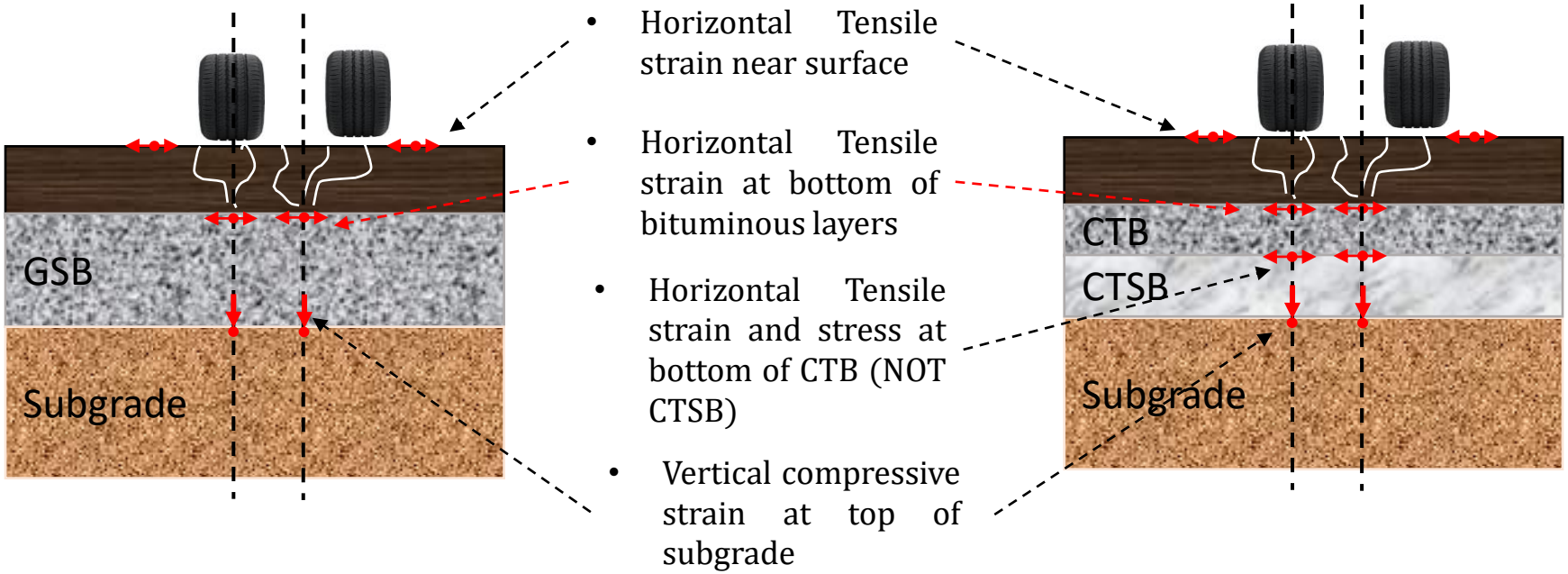
Material Type	M_R/Elastic Modulus (E)	Poisson's Ratio (μ)
Bituminous Layer with VG 40	3000	0.35
Bituminous Layer with VG 30	2000	0.35
Cement Treated Base	5000	0.25
Cement Treated subbase	600	0.25
Cold recycled base	800	0.35
Granular interlayer	450	0.35
Unbound Granular Layers	$0.2\ (h)^{0.45} \times M_{RS}$	0.35
Unbound Granular base over CTSB	300 for natural gravel 350 for crushed aggregates	0.35
Subgrade	$10.0 \times CBR$ for $CBR \leq 5\ \%$ $17.6 \times (CBR)^{0.64}$ for $CBR > 5\ \%$	0.35

Response Model: Layered Elastic Model

❖ Analysis of Stresses in Flexible Pavements using Layered Elastic model

- Under the load application (circular loading with radius 'a'), following responses will be obtained

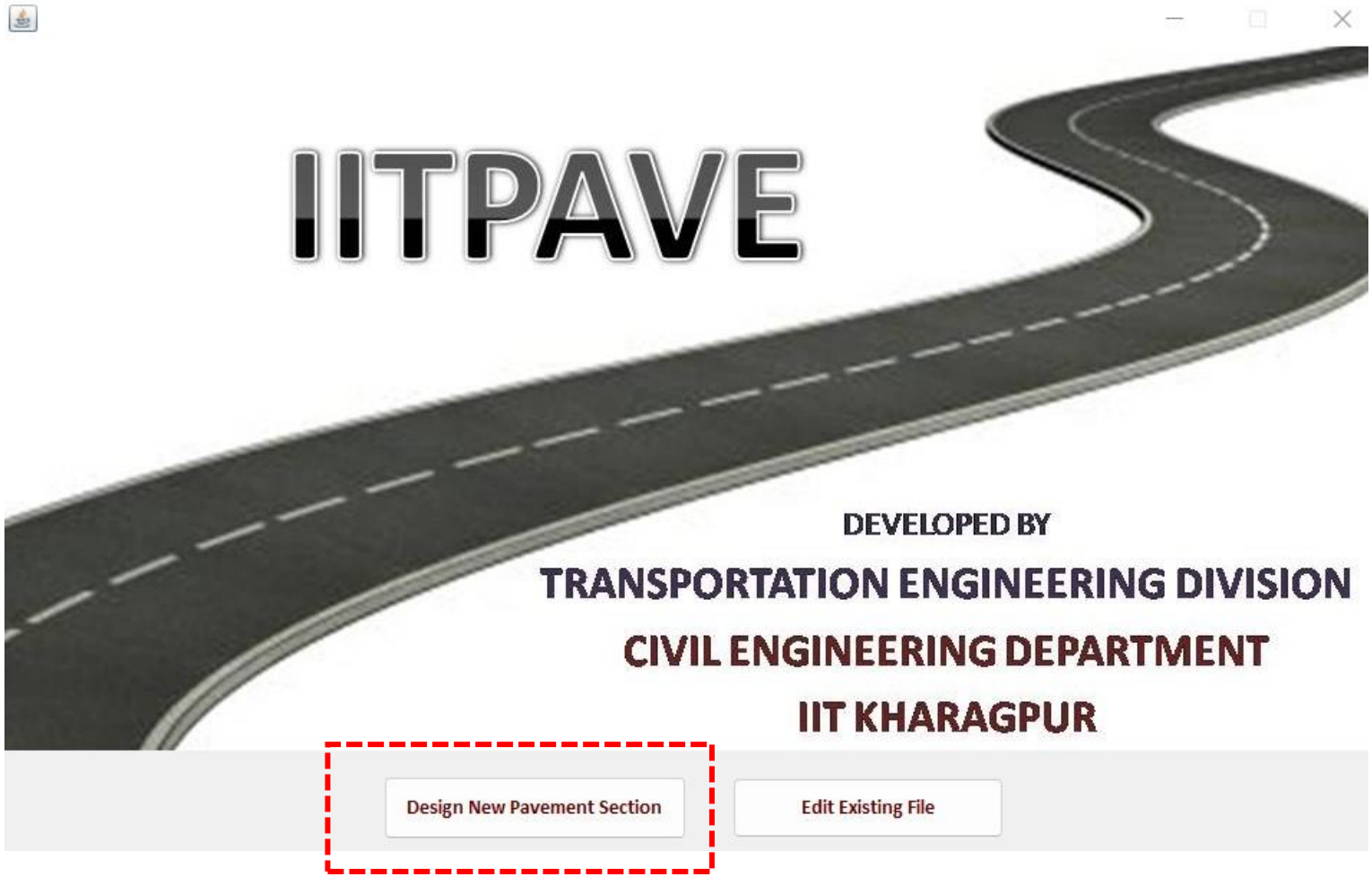
i. Vertical strain on top of subgrade
ii. Horizontal Tensile strain between different layers due to friction (strain) and at the surface of top layer
iii. Radial & vertical stresses



- Horizontal Tensile strain near surface
- Horizontal Tensile strain at bottom of bituminous layers
- Horizontal Tensile strain and stress at bottom of CTB (NOT CTSB)
- Vertical compressive strain at top of subgrade

- There are difficult tables which can be used to predict the responses for different E, a and P values; however, we can use software (IITPAVE, KENPAVE) to obtain the responses easily and quickly

Response Model: IITPAVE



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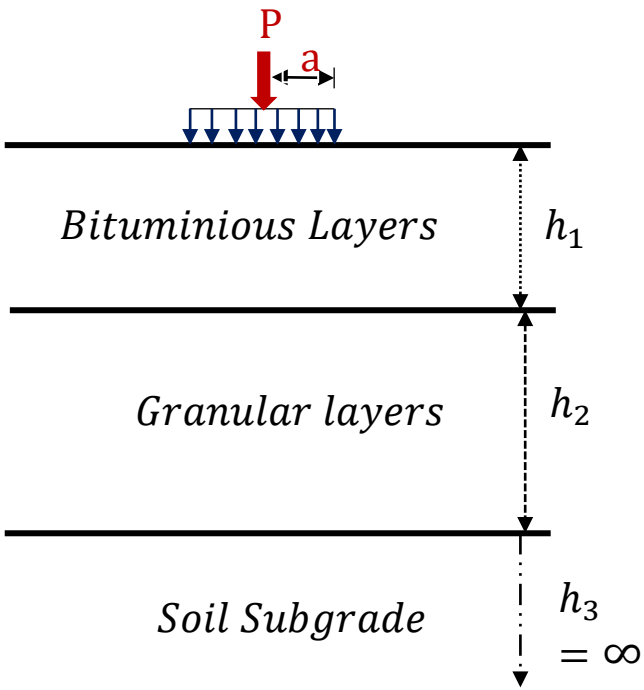
❖ Structural Model: What should be components ? How will the system response?



Parameters affecting the design

- What should be Load? Adjusted Standard loads using VDF
- Position of Load? Lane distribution factor
- Contact Pressure? 0.56 MPa
- Strength (E and μ)
- Response (vertical stress, radial strain (friction) and radial stress)

❖ Now lets see what are the critical responses under which the pavement fails

Mechanical-Empirical Design Procedure (MEPD): Steps

Structural Model

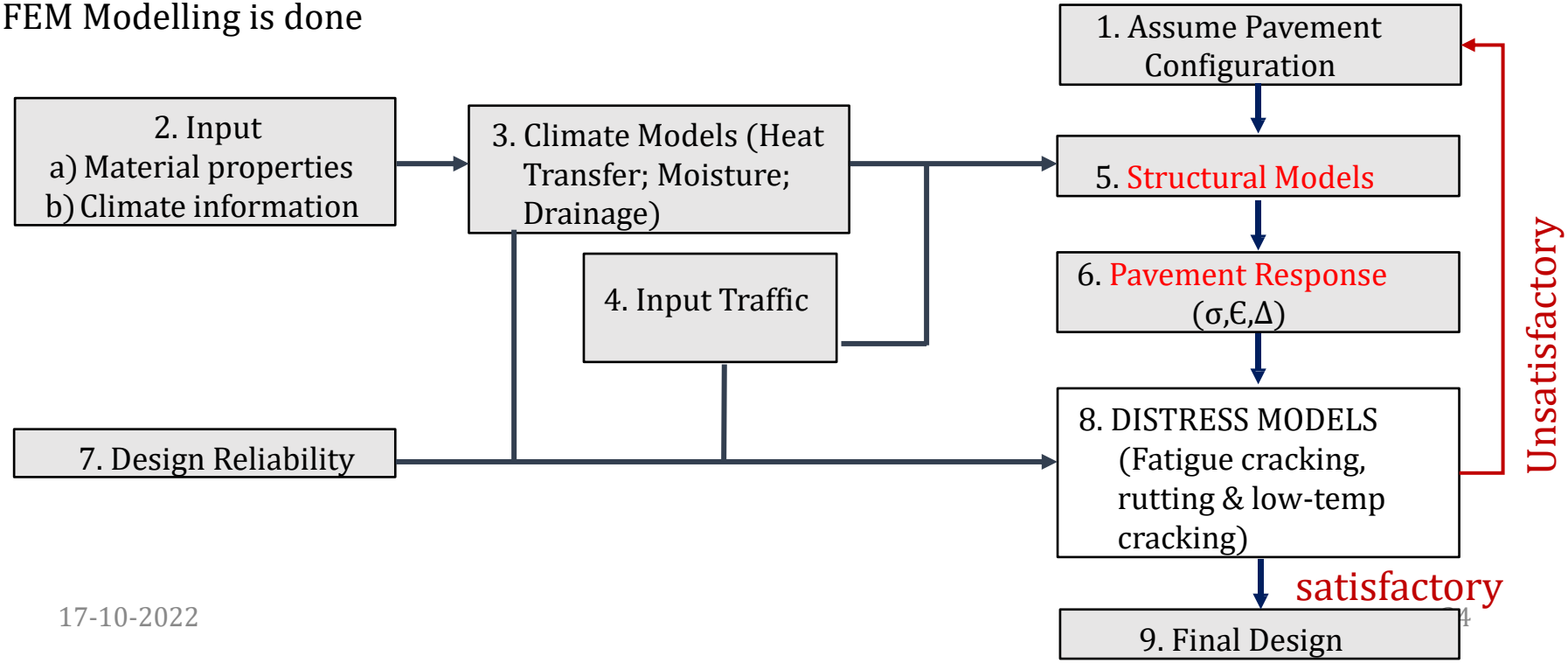
- Design the structure and predict the pavement response (stress & strain; deflection) with different input parameters
- FEM Modelling is done

Distress Models

- Predict the distress (failure) from pavement response
- Empirical regression eqn

Calibration of Prediction with Field

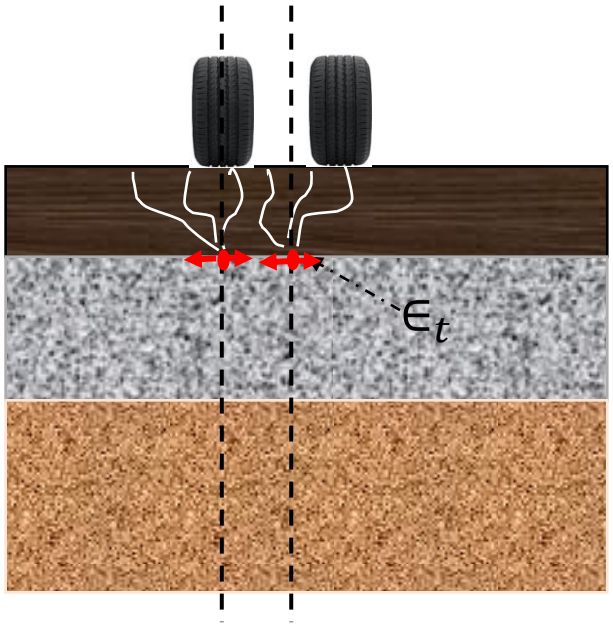
- Calibrate the predicted distress with the field distress



Distress Models

❖ Critical Failure Criterions considered in IRC:37-2018

1. Fatigue Cracking due to horizontal tensile strain (ϵ_t) at the bottom of bituminous layers



➢ Important Points to be noted:

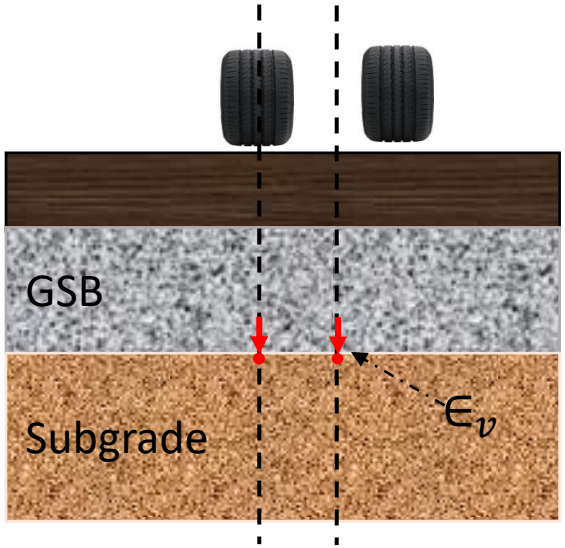
- A pavement is said to be failed if the cracking due to fatigue (interconnected cracks as shown in above figure) is ≥20% of the total surface areas.

- A pavement is said to be SAFE if the actual/expected tensile strain (ϵ_t) at the bottom of the bituminous layers calculated through IITPAVE is lesser than allowable tensile strain (ϵ_t) that leads to fatigue cracking

Distress Models

❖ Critical Failure Criterions considered in IRC:37-2018

2. Rutting due to vertical compressive strain ($\in_t$) over the subgrade



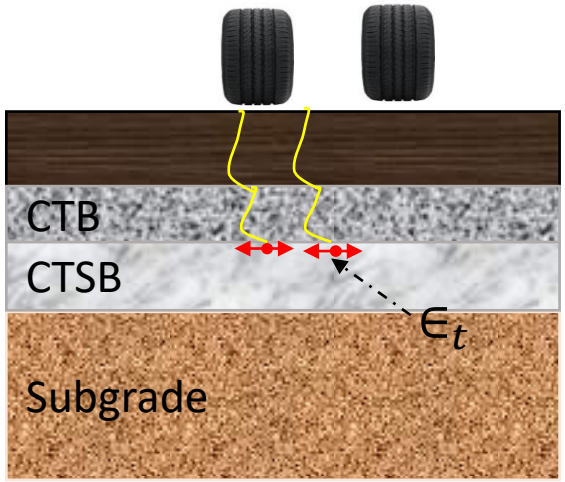
➢ Important Points to be noted:

- A pavement is said to be failed if the rut depth is ≥20 mm (as indicated by the rut reflection on the surface layer)
- A pavement is said to be SAFE if the actual/expected compressive strain ($\in_v$) at the top of the subgrade computed through IITPAVE is lesser than allowable compressive strain ($\in_v$) that leads to fatigue cracking

Distress Models

❖ Critical Failure Criterions considered in IRC:37-2018

3. Fatigue of the Cement Treated Base due to tensile strain (ϵ_t) below CTB



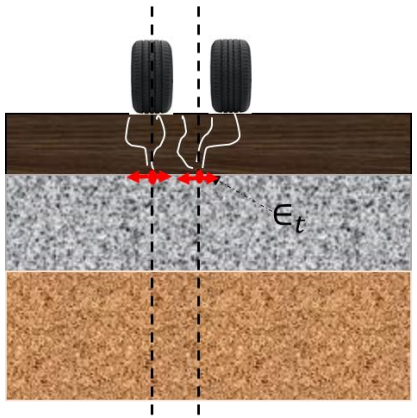
➢ Important Points to be noted:

- A pavement is said to be failed if it develops fatigue cracks (transverse and longitudional) which then reflects on the surface of bituminous surface
- A pavement is said to be SAFE if the actual/expected tensile strain (ϵ_t) at the bottom of the CTB computed through IITPAVE is lesser than allowable tensile strain (ϵ_t) that leads to fatigue cracking
- In addition to the above criterion, cumulative fatigue damage (CFD) similar to rigid pavement shall be less than 1 for all the axles (single, tandem and tridem) and loading conditions

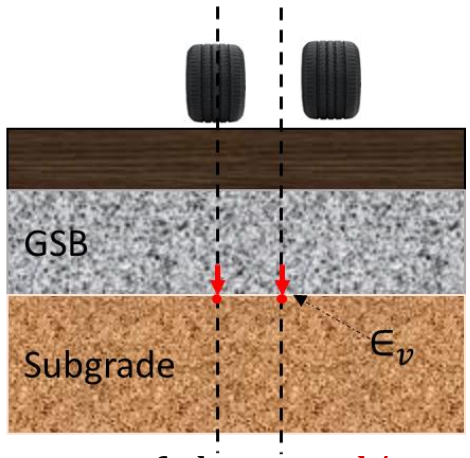
Distress Models

❖ Summary of the Distress Models

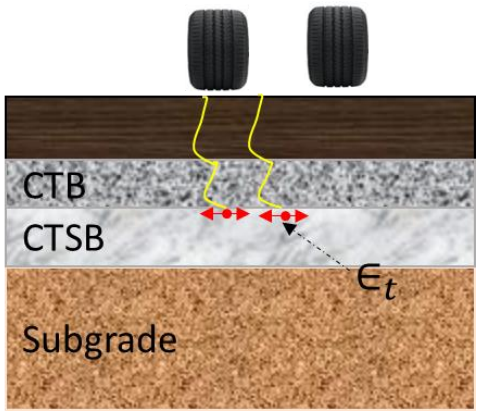
➢ A pavement is said to be SAFE, if



Fatigue: Actual/expected tensile strain (ϵ_t) at the bottom of the bituminous layers computed through IITPAVE is lesser than allowable tensile strain (ϵ_t) that leads to fatigue cracking



Rutting: if the actual/expected compressive strain (ϵ_v) at the top of the subgrade computed through IITPAVE is lesser than allowable compressive strain (ϵ_v) that leads to fatigue cracking



CTB: if the actual/expected tensile strain (ϵ_t) at the bottom of the CTB computed through IITPAVE is lesser than allowable tensile strain (ϵ_t) that leads to fatigue cracking

- How to Expected/Allowable tensile/compressive strain (ϵ_t) (ϵ_v) ?

Fatigue Model

1. Steps to determine the Allowable strain values

2.1 Allowable tensile strain (ε_t)at bottom of bituminous layers for Fatigue

For low-volume roads (< 20 MSA) (for 80 % reliability)

$$N = 1.6064 \times C \times 10^{-4} \times \left(\frac{1}{\varepsilon_t}\right)^{3.89} \times \left(\frac{1}{M_R}\right)^{0.854}$$

For Expressway/NH/SH/urban roads (for 90 % reliability)

$$N = 0.5161 \times C \times 10^{-4} \times \left(\frac{1}{\varepsilon_t}\right)^{3.89} \times \left(\frac{1}{M_R}\right)^{0.854}$$

- N= expected load repetitions
- C = Adjustment Factor to take into the account the effect of mix volumetrics on the fatigue performance of asphalt concrete

$$C = 10^{M} \qquad \mathrm{M} = 4.84\left[\frac{V_{be}}{V_a + V_{be}} - 0.69\right]$$

- V_{be}= Effective bitumen content
- V_a = air void content

- M_R= Resilient Modulus (MPa) of the bituminous mix

Response of Structure using IITPAVE

❖ Resilient Modulus & Poisson's ratio

Material Type	M_R/Elastic Modulus (E)	Poisson's Ratio (μ)
Bituminous Layer with VG 40	3000	0.35
Bituminous Layer with VG 30	2000	0.35
Cement Treated Base	5000	0.25
Cement Treated subbase	600	0.25
Cold recycled base	800	0.35
Granular interlayer	450	0.35
Unbound Granular Layers	$0.2\ (h)^{0.45} \times M_{RS}$	0.35
Unbound Granular base over CTSB	300 for natural gravel 350 for crushed aggregates	0.35
Subgrade	$10.0 \times CBR$ for $CBR \leq 5\ \%$ $17.6 \times (CBR)^{0.64}$ for $CBR > 5\ \%$	0.35

Fatigue Model

Expected Load repetitions (N_f)

$$N = \frac{365 \times A \times [(1+r)^n - 1]}{r} \times LDF \times VDF$$

- N= expected load repetitions in the design years
  - it shall be represented in terms of Millions (i.e. $\frac{1}{10^6}$) – *Million Standard Axles (MSA)*
- r= rate of traffic growth (*maybe assumed as 5%)*
- n= design life, years (*20 years for Expressway/NH/SH & 15 years for other roads*)
- A = initial traffic in the year of completion of construction
  - $A = P\,(1+r)^n$ where P & A are traffic before and after construction

- VDF= Vehicle Damage Factor

CVPD (2-way)	Terrain	
	Rolling/Plain	Hilly
0-150	1.7	0.6
150-1500	3.9	1.7
>1500	5.0	2.8

- LDF = Lane distribution factor

Undivided Carriageways/Roads

Total Lanes	D
Single Lane Roads	100%
2 Lane Single Carriageway	50%
4 Lane Single Carriageway	40%

Divided Carriageways/Roads

Lanes in each direction	D
Two lanes	75%
Three Lanes	60%
Four Lanes	45%

Fatigue Model

2. Steps to determine the Allowable strain values

2.2 Allowable vertical compressive strain (ε_v)at top of subgrade layer for Rutting

For low-volume roads (< 20 MSA) (for 80 % reliability)

$$N = 4.1656 \times 10^{-8} \times \left(\frac{1}{\varepsilon_v}\right)^{4.5337}$$

For Expressway/NH/SH (>20 MSA) (for 90 % reliability)

$$N = 1.41 \times 10^{-8} \times \left(\frac{1}{\varepsilon_v}\right)^{4.5337}$$

2.3 Allowable vertical tensile strain (ε_t)at bottom of CTB for fatigue

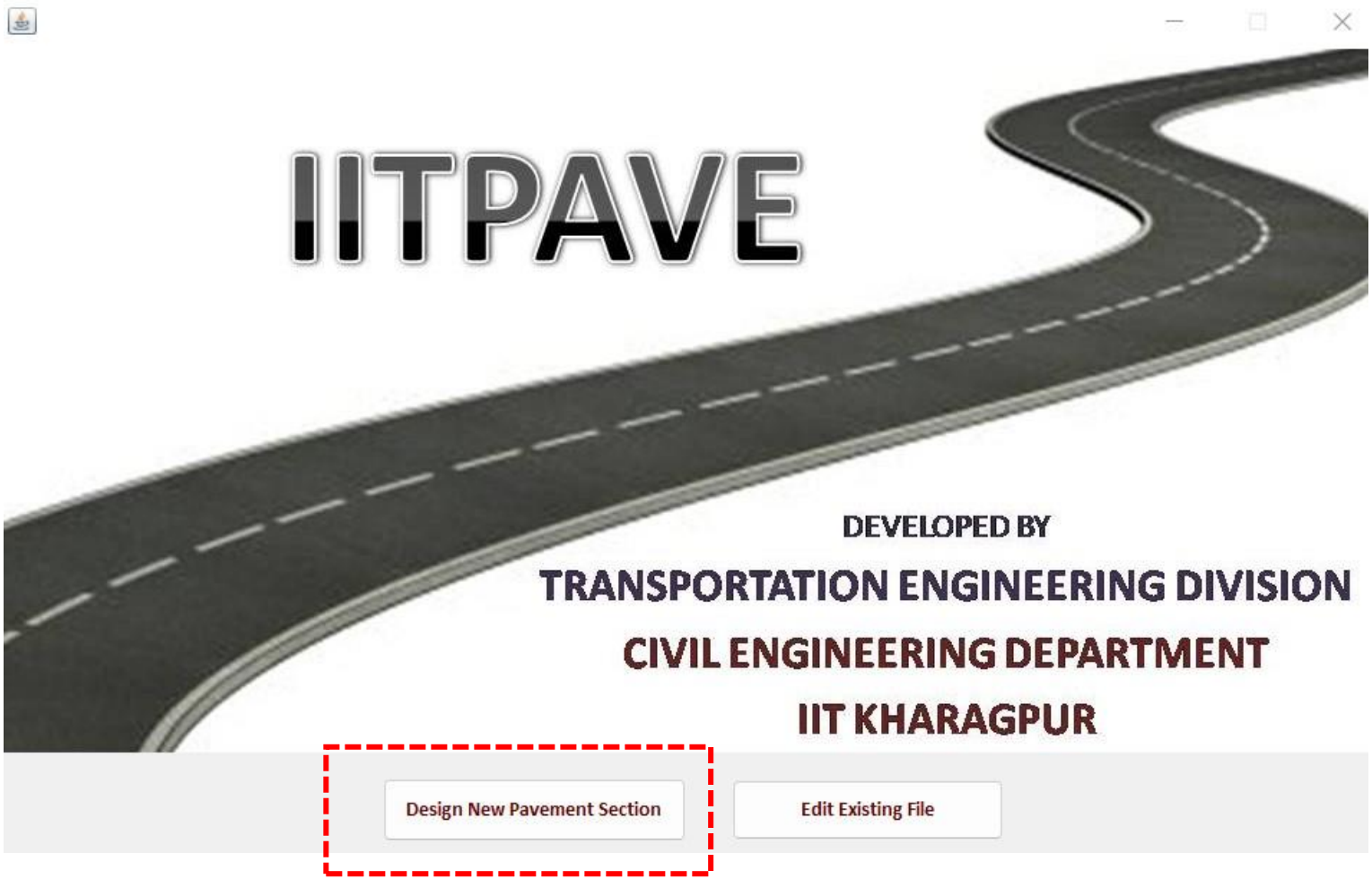
$$N = \left[\frac{\frac{113000}{E^{0.804}} + 191}{\epsilon_t}\right]^{12}$$

- N_f = expected load repetitions
- E = elastic modulus of CTB

Fatigue Model

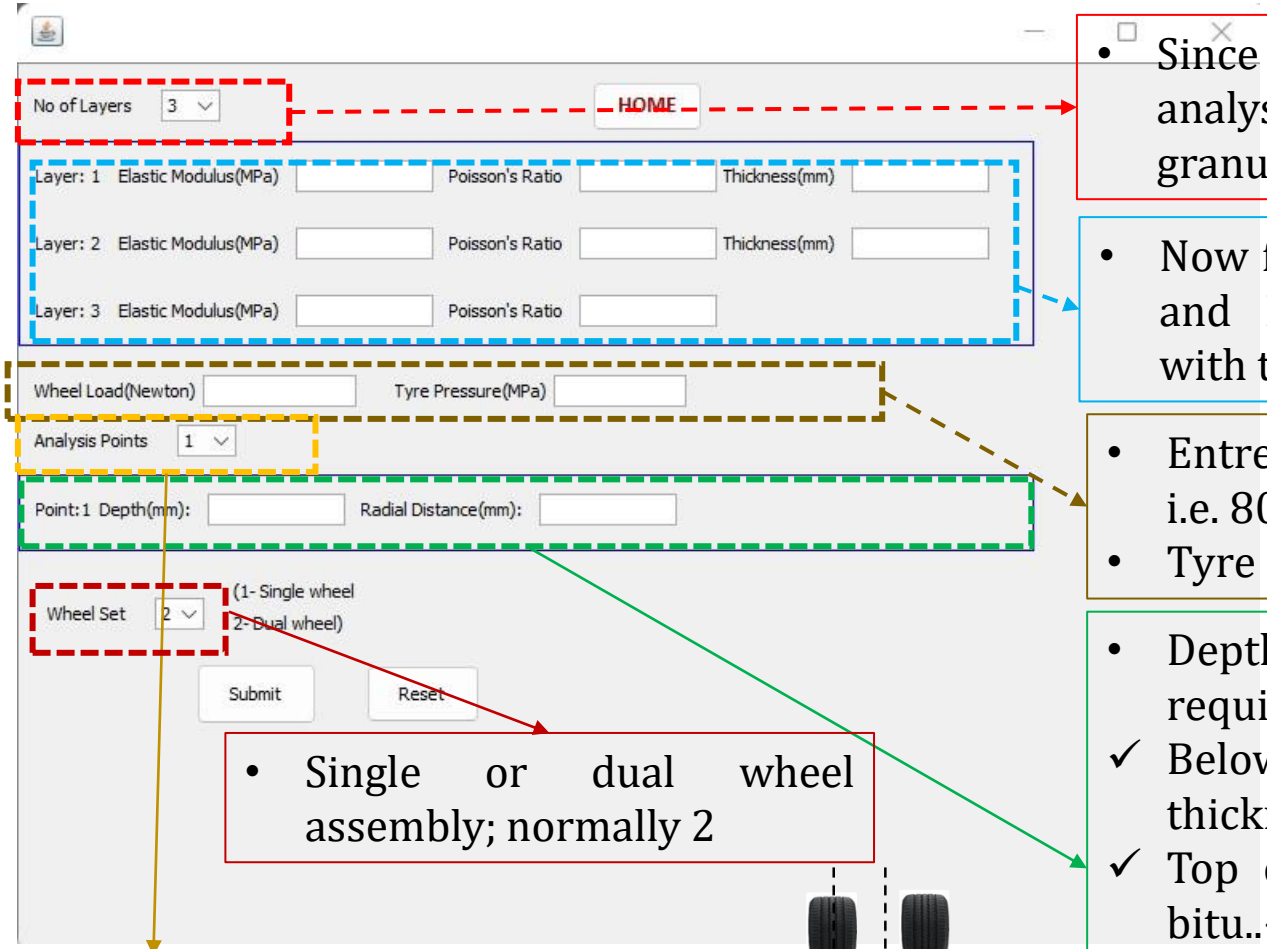
3. Steps to determine the Actual/Expected strain values

2.2 We have to use IITPAVE Software for this



Response Model: IITPAVE

Step 5: Calculate the allowable strain values using IITPAVE Software



- Since we have three layers for analysis, namely, bituminous, granular and subgrade, entre 3

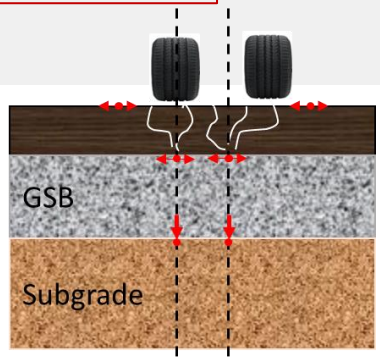
- Now for each layer, input the M_R/E and Poisson's ratio value along with the thickness

- Entre load of one wheel i.e. 80,000 N/4 = 20,000 N
- Tyre Pressure = 0.56 MPa

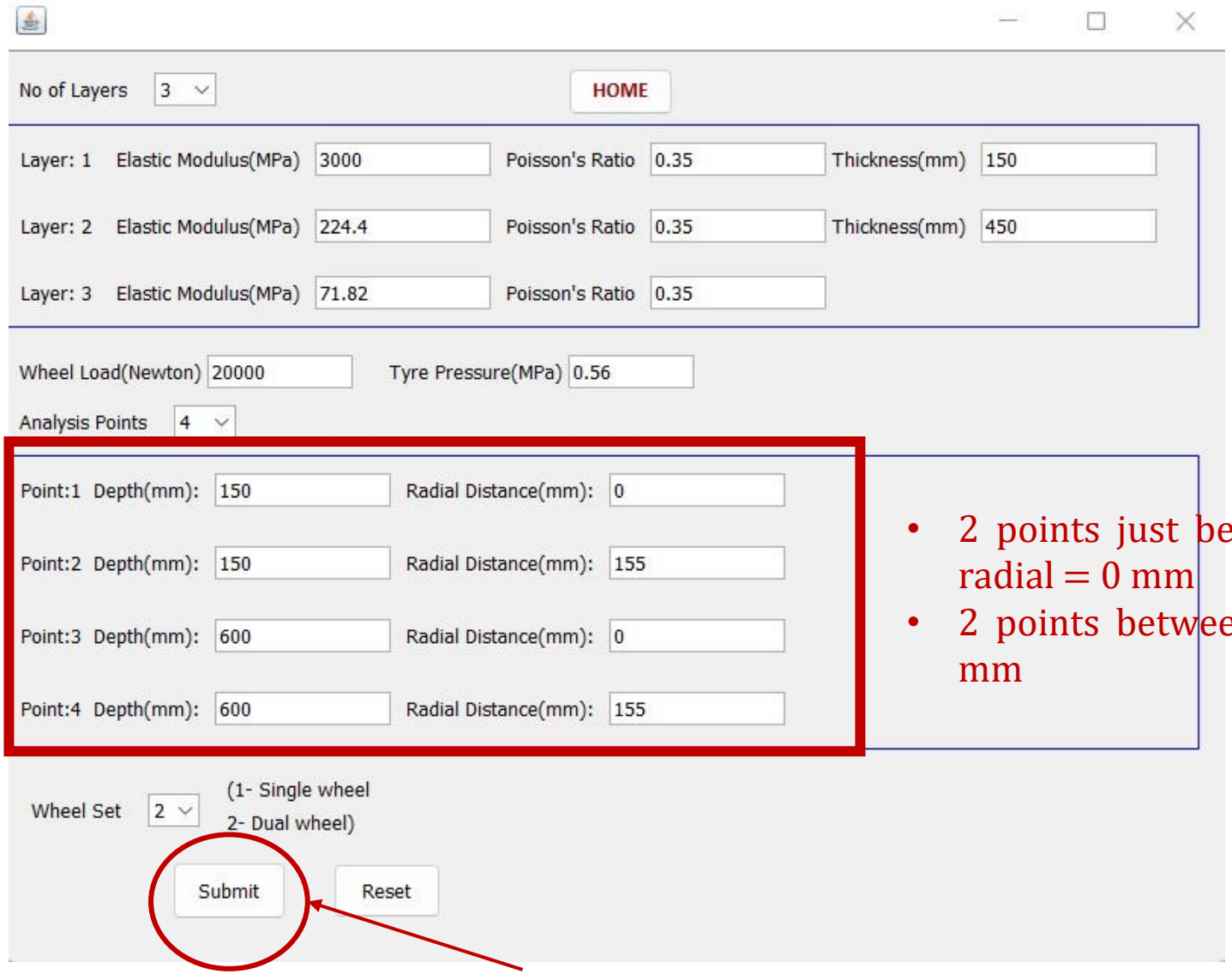
- Single or dual wheel assembly; normally 2

- Depths at which analysis is required
  - ✓ Below bituminous layer i.e. thickness of bituminous
  - ✓ Top of subgrade i.e. thickness of bitu..+granular layer
  - ✓ Radial distance = half of c/c spacing between tires = 310/2 = 155 mm

- Analysis points are usually 4 i.e. 2 points just below the wheels and 2 radially between the tires



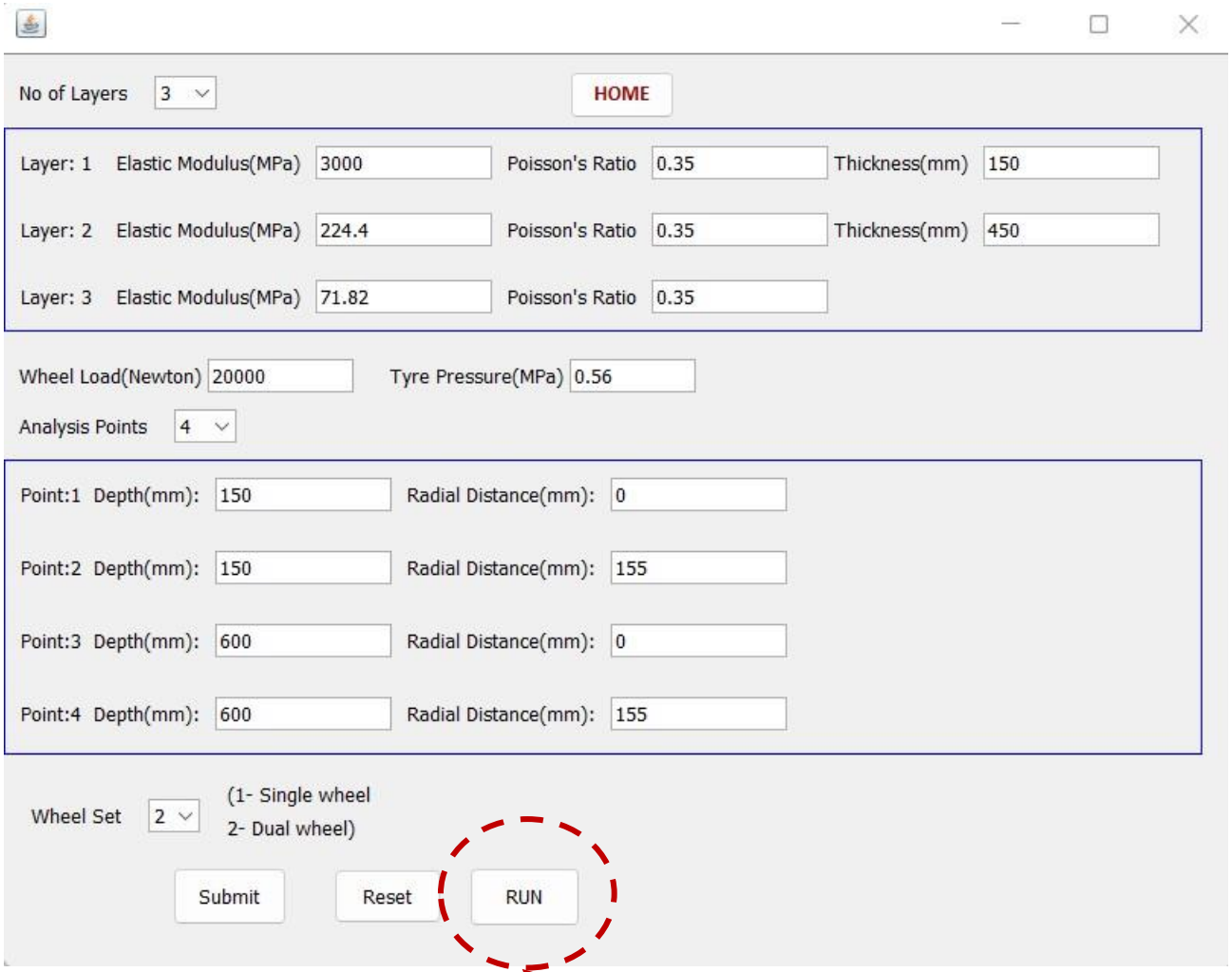
Response Model: IITPAVE



- 2 points just below the wheel i..e radial = 0 mm
- 2 points between the tires = 155 mm

- Click on submit

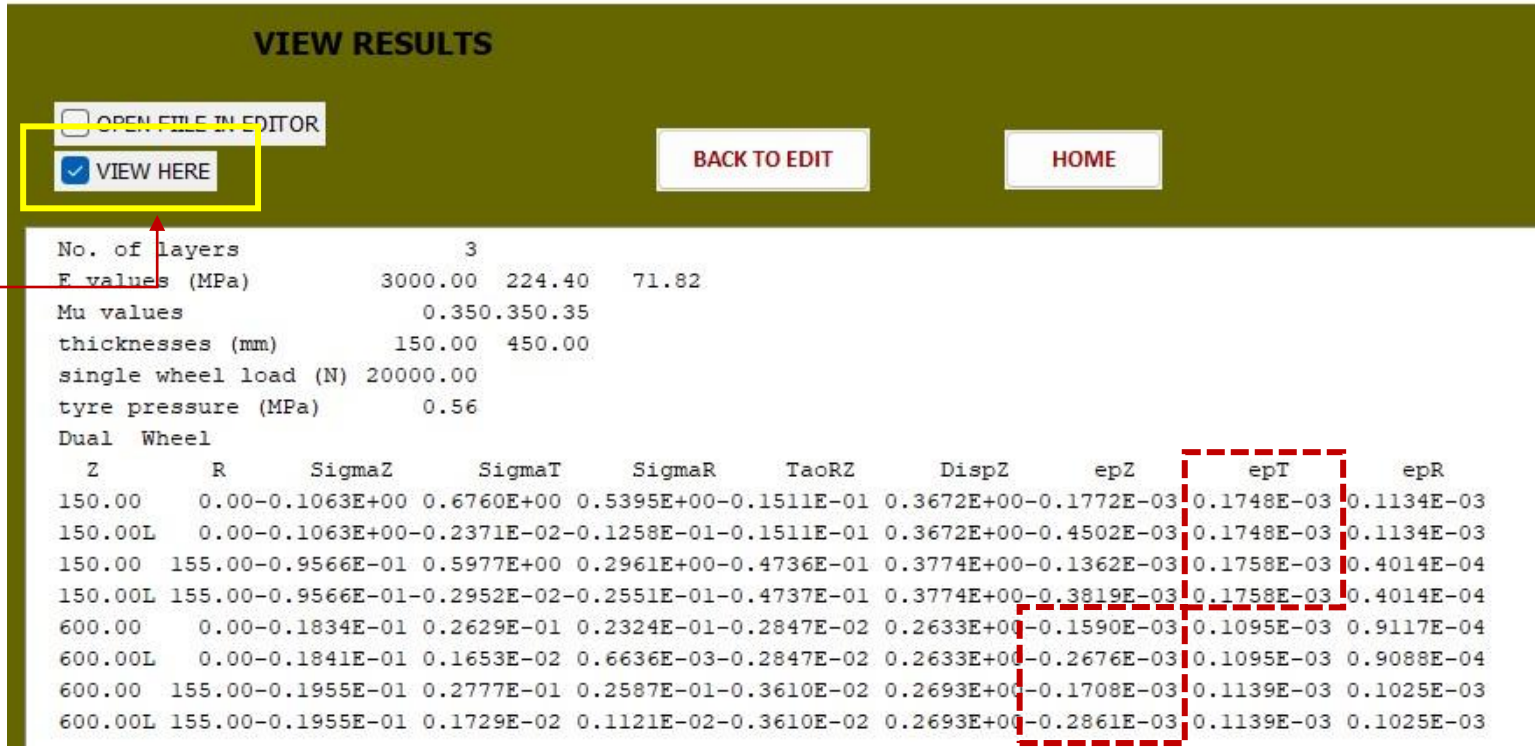
Response Model: IITPAVE



- Click on RUN

Response Model: IITPAVE

- Click on VIEW HERE



No. of layers 3
E values (MPa) 3000.00 224.40 71.82
Mu values 0.350.350.35
thicknesses (mm) 150.00 450.00
single wheel load (N) 20000.00
tyre pressure (MPa) 0.56
Dual Wheel

Z	R	SigmaZ	SigmaT	SigmaR	TaoRZ	DispZ	epZ	epT	epR
150.00	0.00	-0.1063E+00	0.6760E+00	0.5395E+00	-0.1511E-01	0.3672E+00	-0.1772E-03	0.1748E-03	0.1134E-03
150.00L	0.00	-0.1063E+00	-0.2371E-02	-0.1258E-01	-0.1511E-01	0.3672E+00	-0.4502E-03	0.1748E-03	0.1134E-03
150.00	155.00	-0.9566E-01	0.5977E+00	0.2961E+00	-0.4736E-01	0.3774E+00	-0.1362E-03	0.1758E-03	0.4014E-04
150.00L	155.00	-0.9566E-01	-0.2952E-02	-0.2551E-01	-0.4737E-01	0.3774E+00	-0.3819E-03	0.1758E-03	0.4014E-04
600.00	0.00	-0.1834E-01	0.2629E-01	0.2324E-01	-0.2847E-02	0.2633E+00	-0.1590E-03	0.1095E-03	0.9117E-04
600.00L	0.00	-0.1841E-01	0.1653E-02	0.6636E-03	-0.2847E-02	0.2633E+00	-0.2676E-03	0.1095E-03	0.9088E-04
600.00	155.00	-0.1955E-01	0.2777E-01	0.2587E-01	-0.3610E-02	0.2693E+00	-0.1708E-03	0.1139E-03	0.1025E-03
600.00L	155.00	-0.1955E-01	0.1729E-02	0.1121E-02	-0.3610E-02	0.2693E+00	-0.2861E-03	0.1139E-03	0.1025E-03

- epZ = vertical strain and epT = horizontal tensile strain value
- There are 4 each values of epZ and epT:

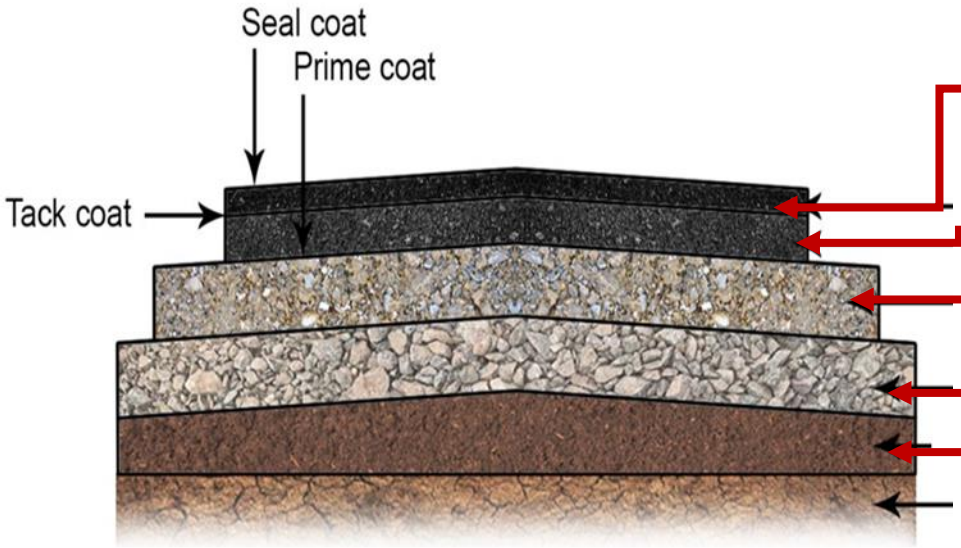
✓ 2 values (e.g. 150 or 600) showing the strain values just above the interface between two layers

✓ 2 values (e.g. 150 L and 600 L) showing the strain values just below the interface between two layers

- See for the maximum values of epT and epZ and these will be the Actual strain values

Fatigue Model

- Some tips for initial thickness calculations



Cross section of typical flexible pavement

Surface Layer = 40 mm

Binder Course= 60-120 mm

Base = WMM = 250 mm
CTB = 100 mm

GSB = 200 mm

Subgrade = 500 mm

- If the criterion fails, increase the thickness in multiples of 5 mm
- If the thickness passes, decrease the thickness to make the design economical

Let's solve some examples

Consider yourself as a highway engineer who have been asked to design of national highway (Four Lane divided carriageway) to be constructed in the State of Andra Pradesh. The following are the data provided to you

- CVPD in both the directions as per last count = 3000
- Construction period = 2 years
- Traffic growth rate per annum = 6%
- Design life = 20 years
- Vehicle Damage Factor = 5.0
- Effective CBR of the subgrade = 9 %
- Mix design parameters of bituminous layer: 3% air voids and 11.5% effective bitumen content

Lets Solve it….

Step 1.0: Calculate the no of CVPD in the year of completion of construction

$$A = P(1 + r)^x$$

$$A = 1500 \times (1 + 0.06)^2$$ Assuming equal traffic in both sides

$$A = 1686 \text{ CVPD}$$

Step 1.1: Calculate the no of CVPD in the design life of 20 years

Let's solve some examples

Step 1.1: Calculate the no of CVPD in the design life of 20 years

$$N = \frac{365 \times A \times [(1+r)^n - 1]}{r} \times LDF \times VDF$$

$$N = \frac{365 \times 1686 \times [(1+0.06)^{20} - 1]}{0.06} \times LDF \times 5.0$$

- LDF = Lane distribution factor

Undivided Carriageways/Roads		Divided Carriageways/Roads	
Total Lanes	D	Lanes in each direction	D
Single Lane Roads	100%	Two lanes	75%
2 Lane Single Carriageway	50%	Three Lanes	60%
4 Lane Single Carriageway	40%	Four Lanes	45%

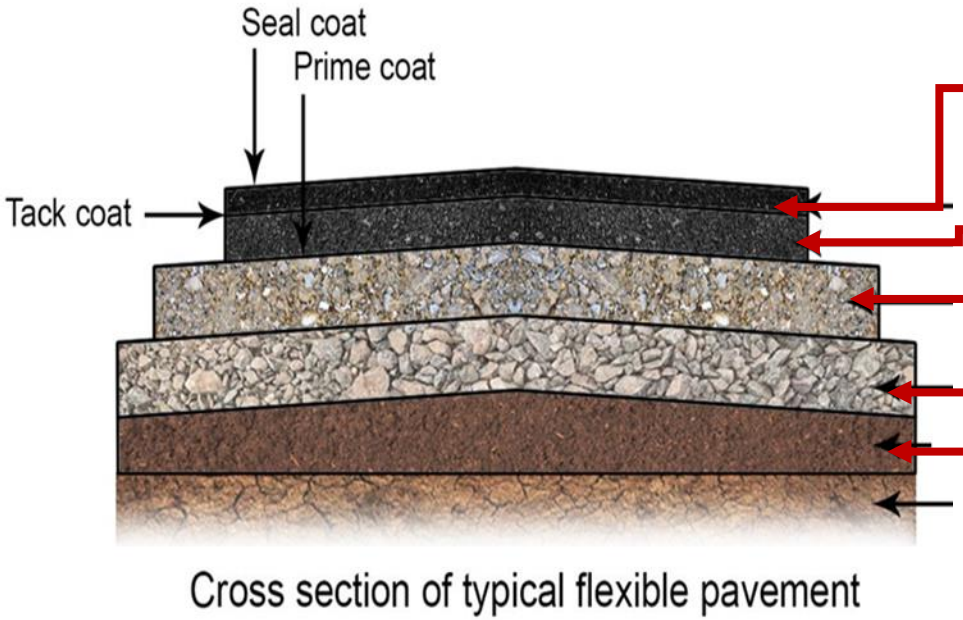
$$N = \frac{365 \times 1686 \times [(1+0.06)^{20} - 1]}{0.06} \times 0.75 \times 5.0$$

$$N = 84890568.64 = 85\ Million\ Standard\ Axles$$

Step 2: Choose the layer combinations

Let's solve some examples

Step 2: Choose the layer combinations



BC (Surface Layer) = 40 mm

Binder Course= 110 mm

Base = WMM = 250 mm

GSB = 200 mm

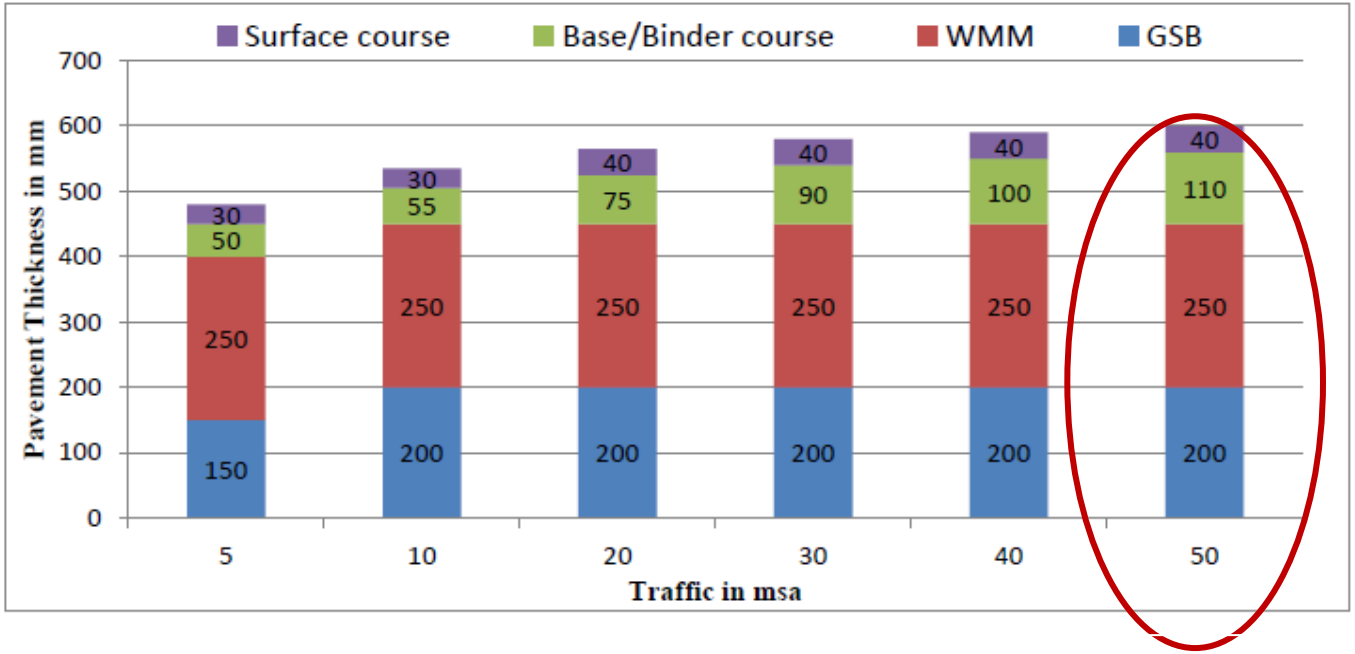
Subgrade = 500 mm

The reason for selecting higher binder thickness is higher traffic (> 50 MSA)

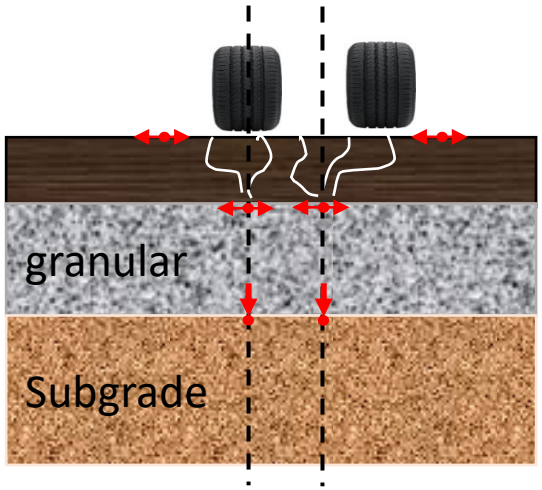
For precious trial thickness, you may refer IRC:37-2018 catalogues

Let's solve some examples

Step 2: Choose the layer combinations (from IRC:37 catalogues)



Step 3: Draw your system boundaries



Bituminous layer= 150 mm

granular layer = 450 mm

subgrade layer = 500 mm

Let's solve some examples

Step 4: Calculate the Allowable strain values

Step 4.1: Calculate the Allowable vertical compressive strain values at top of subgrade for RUTTING Failure

For Expressway/NH/SH (>20 MSA) (for 90 % reliability)

$$N = 1.41 \times 10^{-8} \times \left(\frac{1}{\varepsilon_v}\right)^{4.5337}$$

$$84890569 = 1.41 \times 10^{-8} \times \left(\frac{1}{\varepsilon_v}\right)^{4.5337}$$

$$6.021 \times 10^{15} = \left(\frac{1}{\varepsilon_v}\right)^{4.5337}$$

$$[6.021 \times 10^{15}]^{\frac{1}{4.5337}} = \left(\frac{1}{\varepsilon_v}\right)^{\frac{4.5337}{4.5337}}$$

Dividing both powers by 4.5337

$$\varepsilon_v = \frac{1}{3023.62}$$

$$\varepsilon_v = 3.307 \times 10^{-4}$$

Let's solve some examples

Step 4: Calculate the Allowable strain values

Step 4.2: Calculate the allowable horizontal tensile strain values at bottom of bituminous layer for FATIGUE Failure

For Expressway/NH/SH/urban roads (for 90 % reliability)

$$N = 0.5161 \times C \times 10^{-4} \times \left(\frac{1}{\varepsilon_t}\right)^{3.89} \times \left(\frac{1}{M_R}\right)^{0.854}$$

$$C = 10^{M}$$

$$\mathrm{M} = 4.84\left[\frac{V_{be}}{V_a + V_{be}} - 0.69\right]$$

$$\mathrm{M} = 4.84\left[\frac{0.115}{0.03+0.115} - 0.69\right]$$

$$\mathrm{M} = 0.499$$

$$C = 10^{0.499}$$

$$\mathrm{C} = 3.155$$

- Now we have to calculate the M_R values for bituminous layer
- Its better to calculate the M_R for others layers also for next steps

Let's solve some examples

- Resilient Modulus of Subgrade (M_{RS})

$$M_{RS} = 17.6 \times (CBR)^{0.64} \text{ for CBR} > 5\ \%$$

$$M_{RS} = 17.6 \times (9)^{0.64}$$

$$M_{RS} = 71.82 \text{ MPa}$$

- Resilient Modulus of Granular Layers (M_{RGRAN})

$$MR_{GRAN} = 0.2\ (h)^{0.45} \times M_{RS}$$

$$MR_{GRAN} = 0.2\ (450)^{0.45} \times 71.82$$

$$MR_{GRAN} = 224.4 \text{ MPa}$$

- Resilient Modulus of Bituminous Layers could be selected from the table

Response of Structure using IITPAVE

❖ Resilient Modulus & Poisson's ratio

Material Type	M_R/Elastic Modulus (E)	Poisson's Ratio (μ)
Bituminous Layer with VG 40	3000	0.35
Bituminous Layer with VG 30	2000	0.35
Cement Treated Base	5000	0.25
Cement Treated subbase	600	0.25
Cold recycled base	800	0.35
Granular interlayer	450	0.35
Unbound Granular Layers	$0.2\ (h)^{0.45} \times M_{RS}$	0.35
Unbound Granular base over CTSB	300 for natural gravel 350 for crushed aggregates	0.35
Subgrade	$10.0 \times CBR$ for $CBR \leq 5\ \%$ $17.6 \times (CBR)^{0.64}$ for $CBR > 5\ \%$	0.35

Let's solve some examples

$$N = 0.5161 \times C \times 10^{-4} \times \left(\frac{1}{\varepsilon_t}\right)^{3.89} \times \left(\frac{1}{M_R}\right)^{0.854}$$

$$N = 0.5161 \times 3.155 \times 10^{-4} \times \left(\frac{1}{\varepsilon_t}\right)^{3.89} \times \left(\frac{1}{3000}\right)^{0.854}$$

$$84890569 = 0.5161 \times 3.155 \times 10^{-4} \times \left(\frac{1}{\varepsilon_t}\right)^{3.89} \times \left(\frac{1}{3000}\right)^{0.854}$$

$$84890569 = 0.5161 \times 3.155 \times 10^{-4} \times \left(\frac{1}{\varepsilon_t}\right)^{3.89} \times (1.072 \times 10^{-3})$$

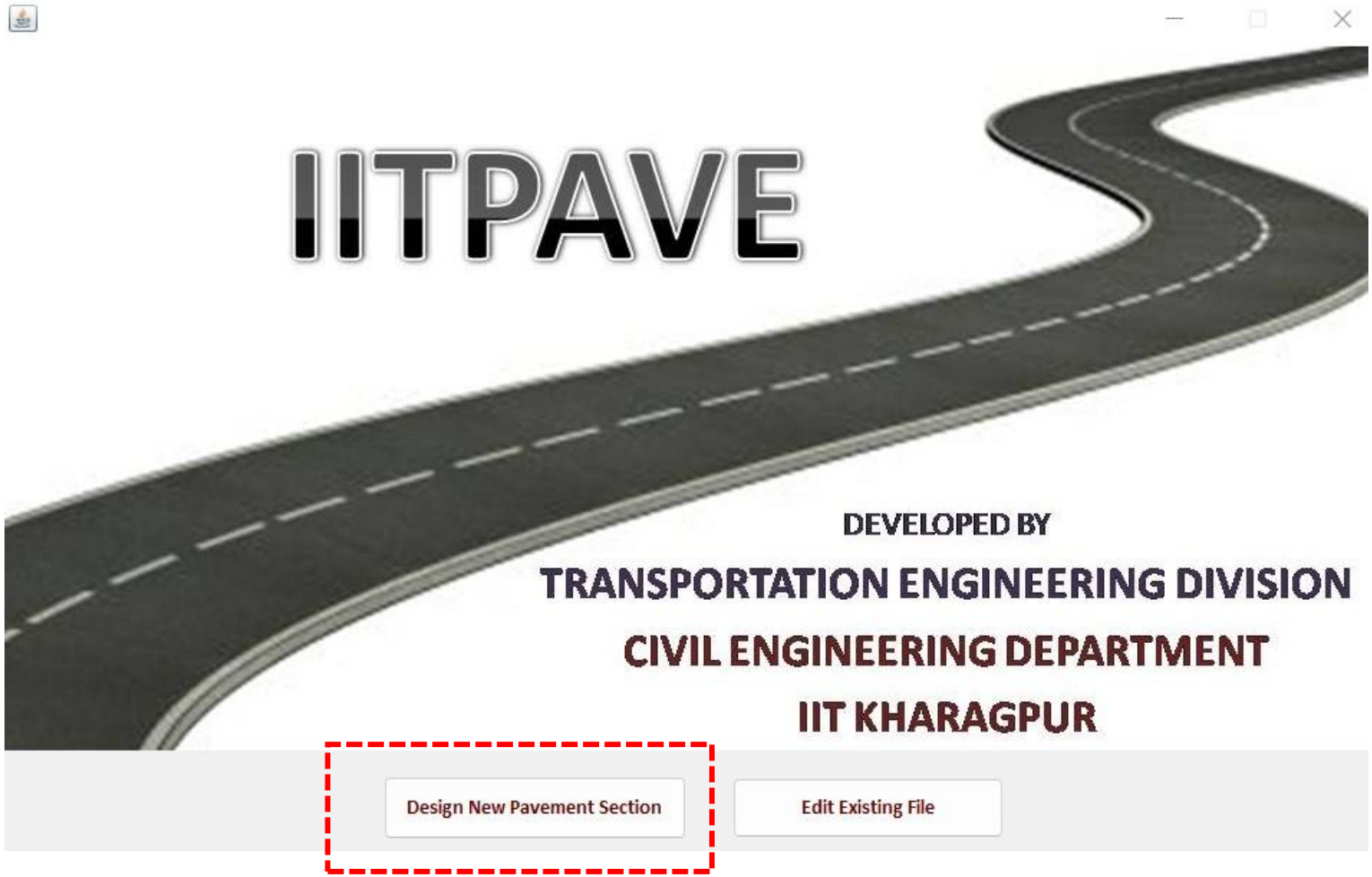
$$4.8633 \times 10^{14} = \left(\frac{1}{\varepsilon_t}\right)^{3.89}$$

$$(4.8633 \times 10^{14})^{\frac{1}{3.89}} = \left(\frac{1}{\varepsilon_t}\right)^{\frac{3.89}{3.89}}$$

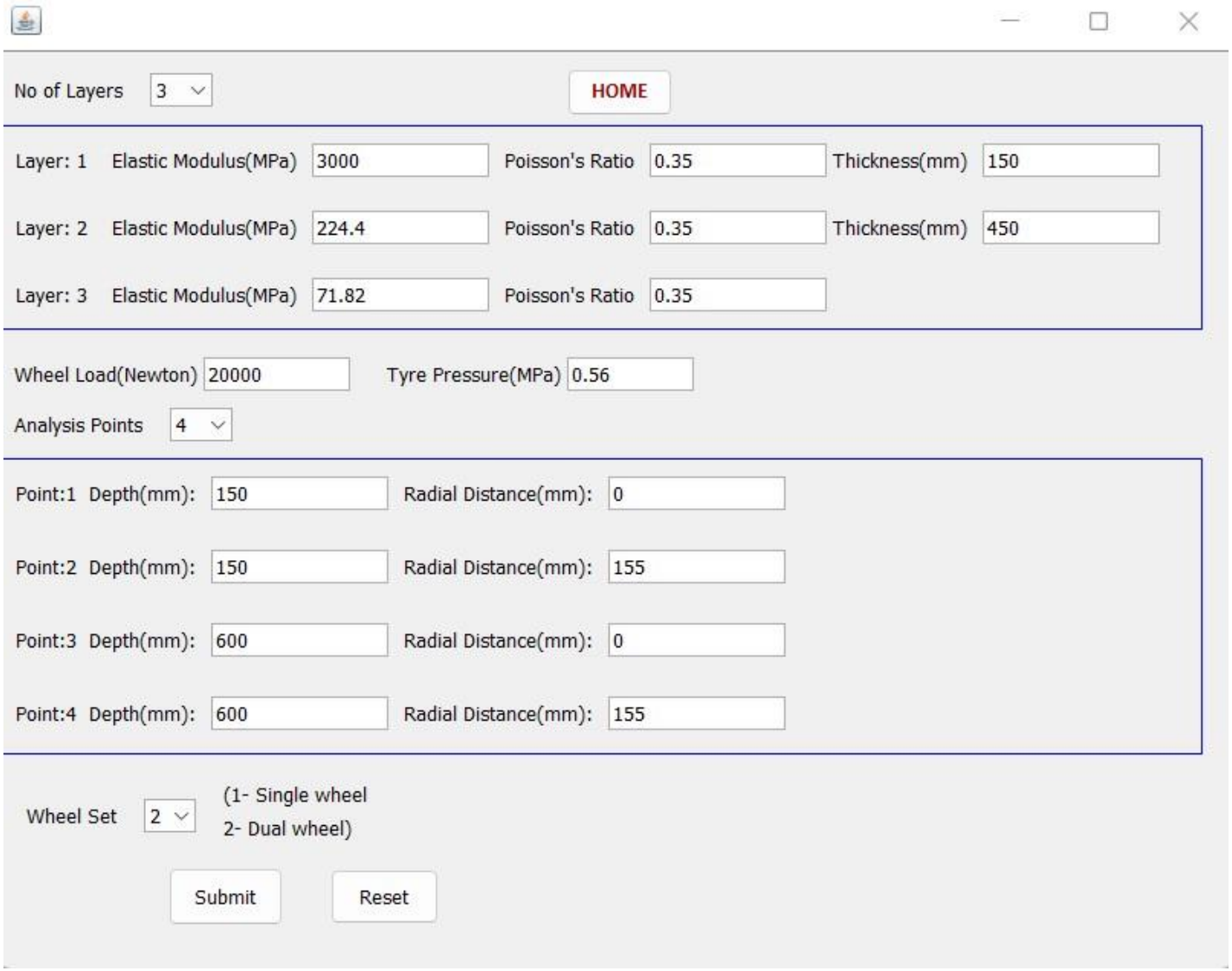
$$\varepsilon_t = \frac{1}{5964.317} \qquad \varepsilon_t = 1.676 \times 10^{-4}$$

Let's solve some examples

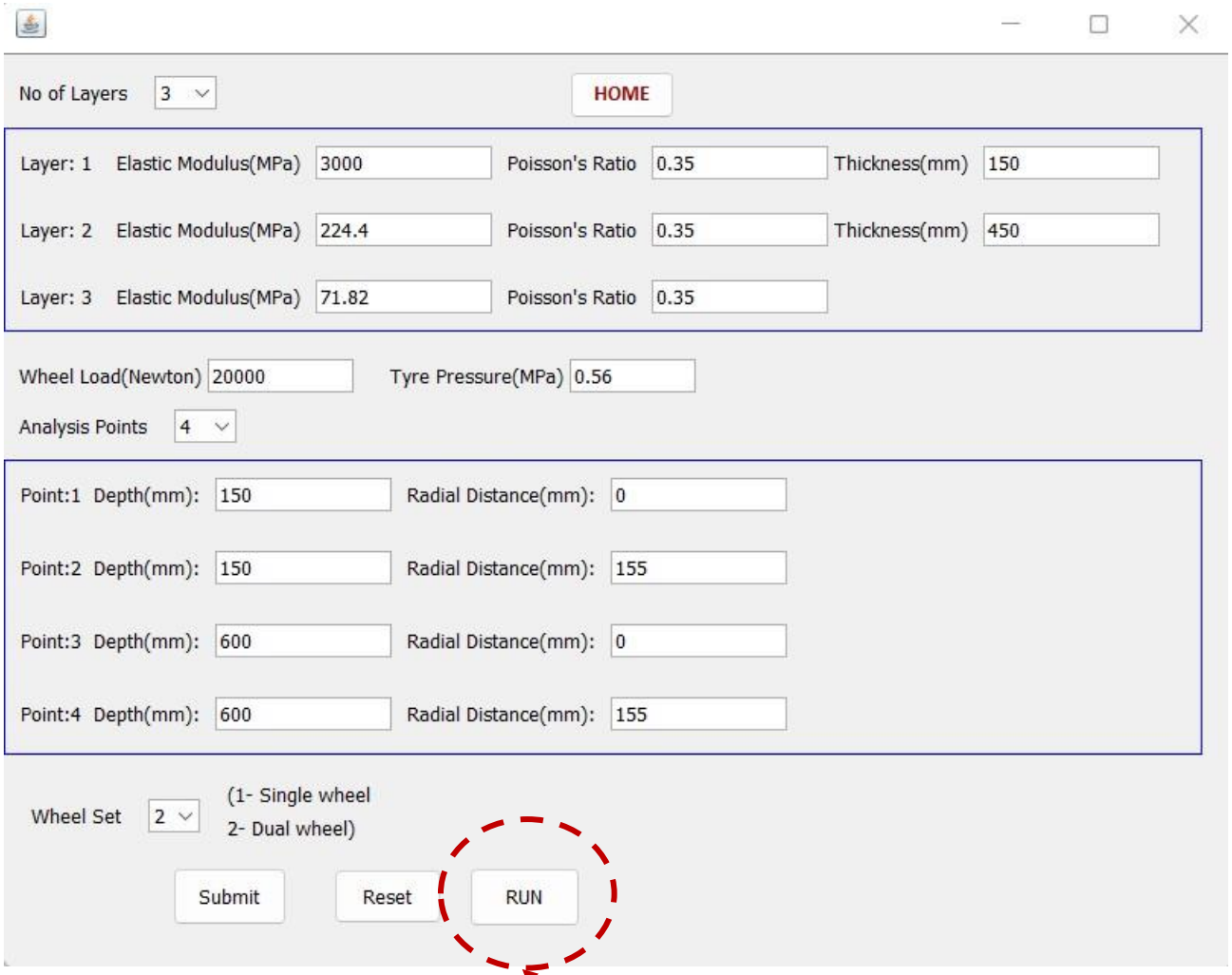
Step 5: Calculate the Actual strain values using IITPAVE Software



Response Model: IITPAVE



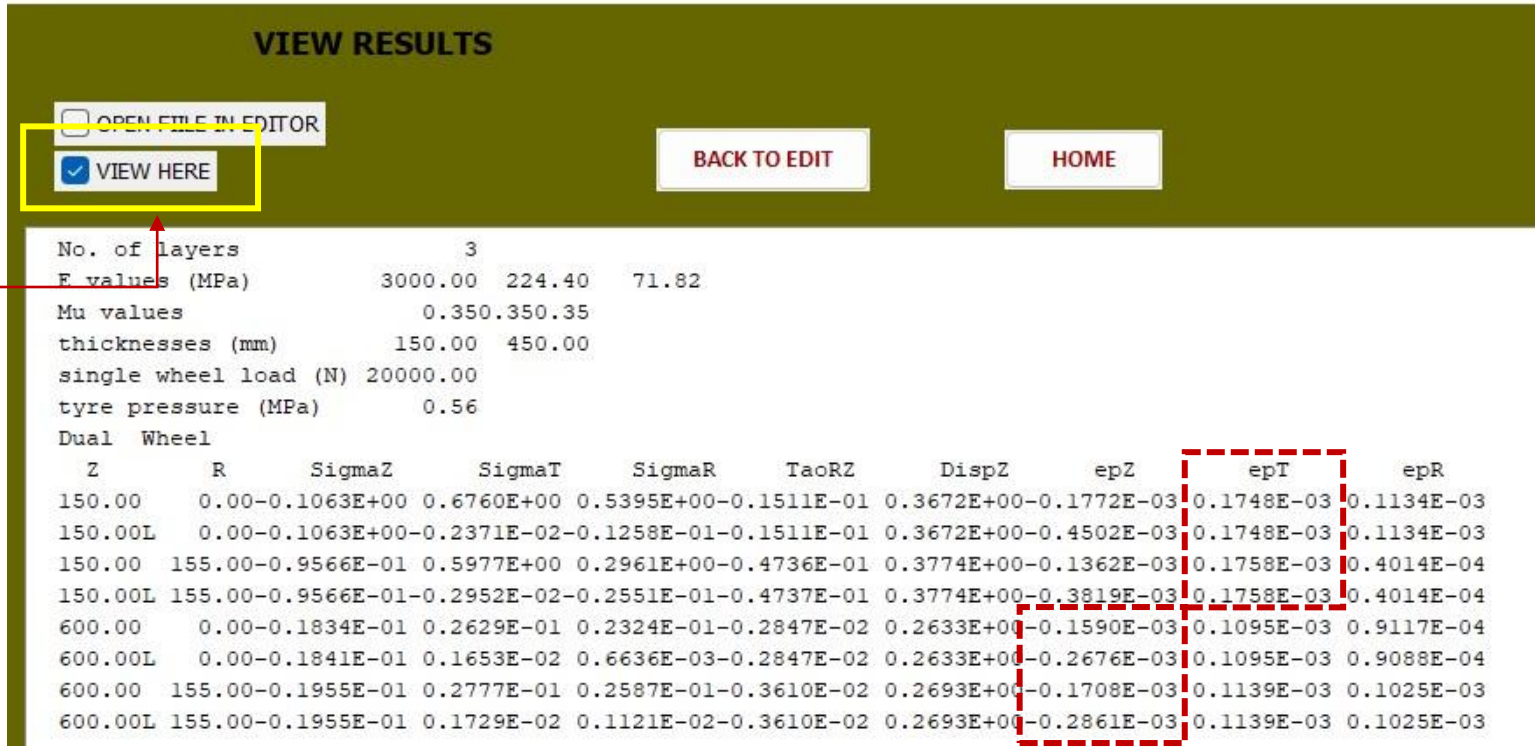
Response Model: IITPAVE



- Click on RUN

Response Model: IITPAVE

- Click on VIEW HERE



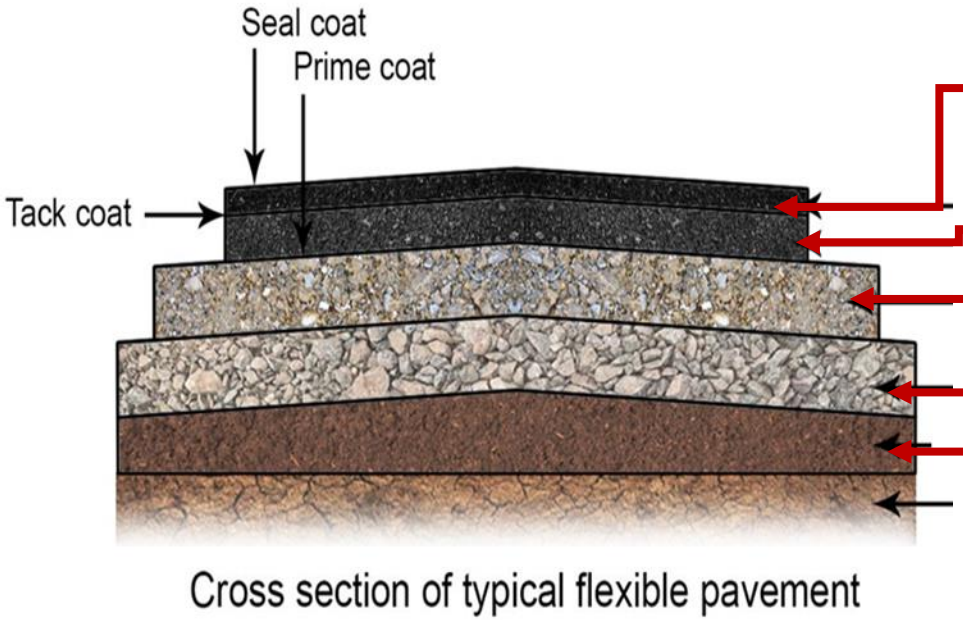

```
No. of layers                 3
E values (MPa)          3000.00  224.40   71.82
Mu values                  0.350.350.35
thicknesses (mm)         150.00  450.00
single wheel load (N) 20000.00
tyre pressure (MPa)        0.56
Dual  Wheel
  Z        R       SigmaZ       SigmaT      SigmaR      TaoRZ       DispZ       epZ        epT        epR
150.00     0.00-0.1063E+00 0.6760E+00 0.5395E+00-0.1511E-01 0.3672E+00-0.1772E-03 0.1748E-03 0.1134E-03
150.00L    0.00-0.1063E+00-0.2371E-02-0.1258E-01-0.1511E-01 0.3672E+00-0.4502E-03 0.1748E-03 0.1134E-03
150.00   155.00-0.9566E-01 0.5977E+00 0.2961E+00-0.4736E-01 0.3774E+00-0.1362E-03 0.1758E-03 0.4014E-04
150.00L 155.00-0.9566E-01-0.2952E-02-0.2551E-01-0.4737E-01 0.3774E+00-0.3819E-03 0.1758E-03 0.4014E-04
600.00     0.00-0.1834E-01 0.2629E-01 0.2324E-01-0.2847E-02 0.2633E+00-0.1590E-03 0.1095E-03 0.9117E-04
600.00L    0.00-0.1841E-01 0.1653E-02 0.6636E-03-0.2847E-02 0.2633E+00-0.2676E-03 0.1095E-03 0.9088E-04
600.00   155.00-0.1955E-01 0.2777E-01 0.2587E-01-0.3610E-02 0.2693E+00-0.1708E-03 0.1139E-03 0.1025E-03
600.00L 155.00-0.1955E-01 0.1729E-02 0.1121E-02-0.3610E-02 0.2693E+00-0.2861E-03 0.1139E-03 0.1025E-03
```

Strain Type	Actual Strain (IITPAVE)	Allowable strain (models)	Remarks
Vertical strain	2.86×10^{-4}	3.307×10^{-4}	Pass
Tensile strain	1.758×10^{-4}	1.676×10^{-4}	Fail

- This means tensile strain below bituminous is higher and therefore, we have to increase the binder layer thickness

Let's solve some examples

Step 2: Choose the layer combinations



BC (Surface Layer) = 40 mm

Binder Course= 120 mm

Base = WMM = 250 mm

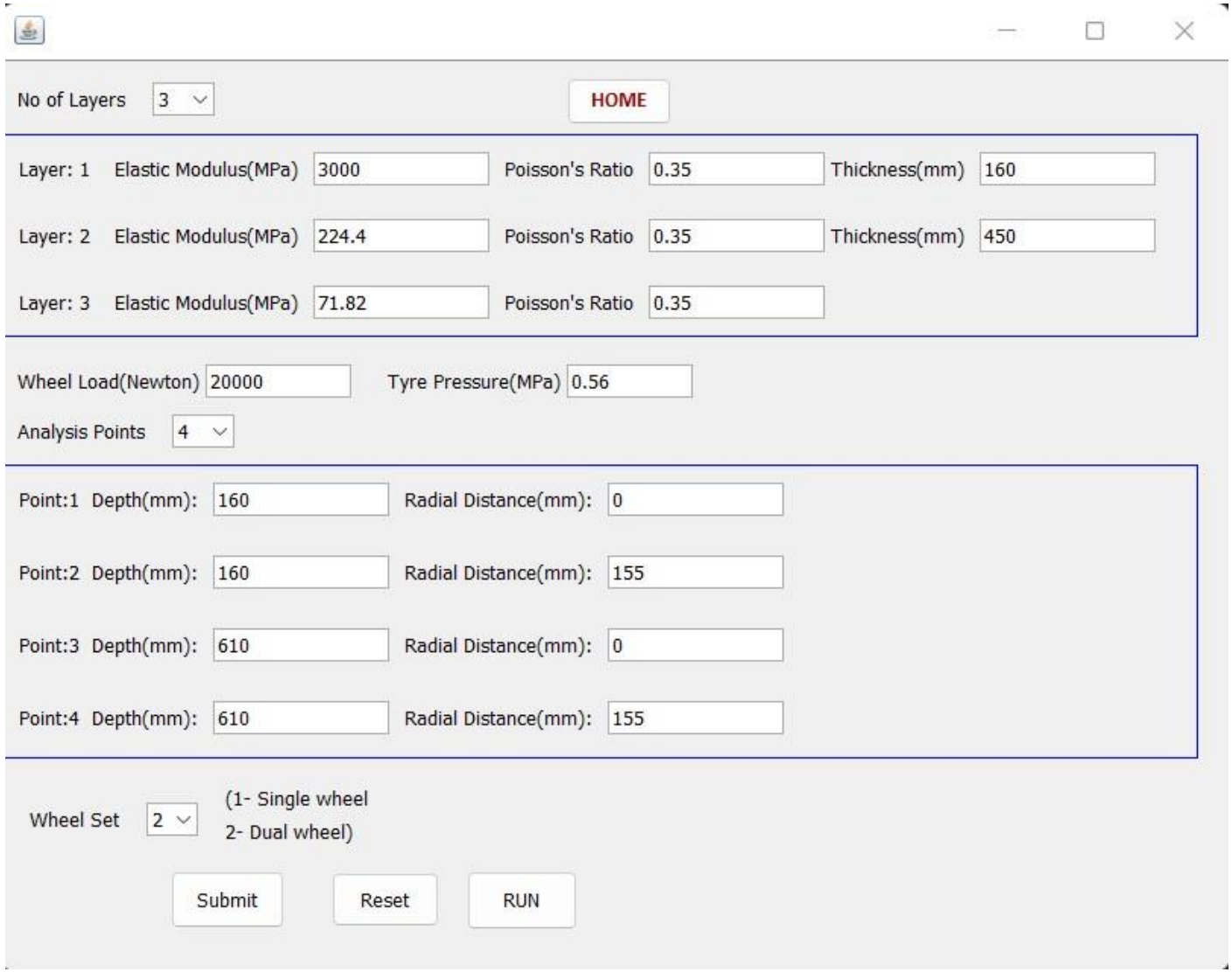
GSB = 200 mm

Subgrade = 500 mm

- The reason for selecting higher binder thickness is higher traffic (> 50 MSA)
- For precious trial thickness, you may refer IRC:37-2018 catalogues

Let's solve some examples

Step 5: Calculate the allowable strain values using IITPAVE Software



Let's solve some examples

Step 5: Calculate the allowable strain values using IITPAVE Software

VIEW RESULTS

OPEN FILE IN EDITOR

VIEW HERE

BACK TO EDIT

HOME

```
No. of layers                3
E values (MPa)          3000.00  224.40   71.82
Mu values                 0.350.350.35
thicknesses (mm)         160.00  450.00
single wheel load (N) 20000.00
tyre pressure (MPa)        0.56
Dual  Wheel
  Z        R      SigmaZ      SigmaT      SigmaR      TaoRZ       DispZ       epZ         epT         epR
160.00     0.00-0.9718E-01 0.6331E+00 0.5060E+00-0.1433E-01 0.3546E+00-0.1653E-03 0.1633E-03 0.1061E-03
160.00L    0.00-0.9718E-01-0.1055E-02-0.1056E-01-0.1433E-01 0.3546E+00-0.4149E-03 0.1633E-03 0.1061E-03
160.00   155.00-0.8913E-01 0.5732E+00 0.3062E+00-0.4314E-01 0.3645E+00-0.1323E-03 0.1657E-03 0.4561E-04
160.00L 155.00-0.8913E-01-0.1529E-02-0.2150E-01-0.4314E-01 0.3645E+00-0.3613E-03 0.1657E-03 0.4561E-04
610.00     0.00-0.1724E-01 0.2492E-01 0.2185E-01-0.2654E-02 0.2573E+00-0.1498E-03 0.1038E-03 0.8541E-04
610.00L    0.00-0.1759E-01 0.1503E-02 0.5917E-03-0.2654E-02 0.2573E+00-0.2552E-03 0.1038E-03 0.8664E-04
610.00   155.00-0.1858E-01 0.2629E-01 0.2457E-01-0.3336E-02 0.2629E+00-0.1621E-03 0.1078E-03 0.9748E-04
610.00L 155.00-0.1858E-01 0.1612E-02 0.1062E-02-0.3335E-02 0.2629E+00-0.2717E-03 0.1078E-03 0.9747E-04
```

Strain Type	Actual Strain (IITPAVE)	Allowable strain (models)	Remarks
Vertical strain	2.71×10^{-4}	3.307×10^{-4}	Pass
Tensile strain	1.657×10^{-4}	1.676×10^{-4}	Pass